Feynman Diagrams for the Variational Free Energy

A primer on variational Laplace, its error term,
and how to compute that term

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Draft — October 20, 2026

Abstract

This is the long-form companion to *Variational Laplace at Two Loops: Feynman Diagrams for the Variational Free Energy* (IWAI 2026), and an expansion of the talk of the same name. It assumes a reader who can write down a variational free energy, find its minimum and invert its Hessian, and who has not met a Feynman diagram.

Variational Laplace reports the first two terms of an asymptotic expansion of the log model evidence. The third term has a closed form in the third and fourth derivatives of the action at the saddle, and is the sum of three Feynman diagrams: the *figure-eight*, the *sunset* and the *spectacles*. Its magnitude estimates the error of the two-term truncation, so a pipeline that computes it reports an evidence together with a bar on that evidence.

The route follows the talk. A single example carries Parts I to III: a Poisson count with a Gaussian prior, in one dimension, where the exact log evidence is available by quadrature and variational Laplace is in error by 0.0104 nats. That error is accounted for in four steps — write the action, read off the Feynman rules, draw the diagrams at the next order, sum them — and the four steps are then repeated with indices restored. Part IV applies the construction to three settings and reports the numbers. Part V covers use, scope and limits. Every numerical claim is reproduced by a named script, and Appendix B translates between the two vocabularies.

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1 How to use this primer

1.1 Assumed background

You can write down a variational free energy, you know what the Kullback–Leibler term in it does, and you have computed a Hessian at a mode and added half its log-determinant to something. Everything else is built here from the one-dimensional Gaussian integral upwards.

No physics is assumed and none is required. The words this subject borrows — action, propagator, vertex, loop, diagram — are introduced below as names for objects assembled from that Gaussian integral, and each is defined where it first appears. Appendix B collects the correspondence with the physics literature for anyone who wants to follow it there, together with a statement of how much of that literature's apparatus is actually in play, which is less than the names suggest. It sits at the end because it presumes the vocabulary built up on the way.

1.2 The example and the four steps

Parts I to III run on one example and one observable.

An agent runs a single trial and counts how many times an outcome occurs. Its belief about the log-rate s is a standard normal, the count is Poisson with rate e^s, and the trial returns $o = 0$. The observable is the log evidence $-\log p(o)$, which is the quantity model comparison consumes. Quadrature gives 0.9630 nats; variational Laplace gives 0.9526; Parts II and III account for the difference.

The accounting has four steps, which are the section titles of Part II:

1. **The action.** Write down what sits in the exponent and expand it about its minimum (§6).
2. **The Feynman rules.** Read one rule off each term of that expansion (§7).
3. **The diagrams.** Combine the rules into every shape that exists at a given order, and evaluate each shape (§8).
4. **The sum.** Add them (§9).

These four steps are the standard perturbative recipe of statistical field theory, presented for example in Negele and Orland [1988, ch. 2], Kardar [2007, ch. 2–3] and Mattuck [1992, ch. 3–4]. Part III repeats them with indices restored.

1.3 The boxes

- **TRANSLATION** gives the physics name for an object defined here in inference terms. The name is the search term for the other literature.
- **QUESTION** answers a question the talk was asked.
- **GOTCHA** records an error that is easy to make and hard to detect. They are collected in §22.
- **WHITEBOARD DRILL** states what you should be able to do unaided at that point.

- **NUMBERS** reports output from a named script, so that a disagreement between your algebra and the text is settled by arithmetic.
- **FURTHER READING** closes each part. These favour books and review articles over research papers, and say what to read, which chapter, and in what order.

Appendix A tabulates every symbol, as a backstop for a reader who has jumped in partway, and Appendix B gives the correspondence with the physics literature.

1.4 Reading orders

Goal	Sections
The idea, in twenty minutes	§2, §4, §9, §18.
Ability to derive the result	Parts I to III in order, with a pen. §9 and §10 reward slowness.
Use on your own model	§21 is the procedure; §10 is the formula; §12 is the error you would otherwise ship.
Knowing when not to trust it	§6.4, §18, §20, §17.
Refereeing it	§12, §16, §23, and the test in §25.

§12 is the section to skip last. It is where the one-dimensional answer stops being correct, and an implementation that has not read it fails in a way that no one-dimensional test detects.

1.5 Exit criteria

1. Do a multivariate Gaussian integral and identify the result as the Occam factor (§7).
2. State Wick's theorem and count the pairings of six objects (§7).
3. Say why the quadratic term of the action is handled differently from the others (§6.2).
4. Derive the two-loop coefficient in one dimension, including both the 1/8 and the 5/24 (§9).
5. Enumerate the 15 pairings of two cubic vertices, split them $6 + 9$, and show that no third topology exists (§10).
6. Say why a one-dimensional test cannot separate 1/12 from 1/8, and design one that can (§12).
7. Write variational Laplace as the first two terms of the series, and point at the diagram that is the second (§11).
8. Give the three readings of the two-loop magnitude against a decision threshold, and the action to take in each (§18).

1.6 The companion code

Five scripts are published with this document, at `https://github.com/egolabs-ai/iwai26` under `two-loops/code`. The main one is `primer_examples.py`, which contains seven exercises keyed to the sections below. It avoids Monte Carlo wherever exact quadrature is available, so that a disagreement between your algebra and its output is an algebra error rather than sampling noise. The dependencies are `numpy` and `scipy` and nothing else:

`uv sync` then `uv run python primer_examples.py`

Appendix E maps every number in this document to the script that produces it.

2 The whole thing in one page

Bayesian inference requires the model evidence $p(o) = \int p(s, o)\, ds$, an integral over hidden states s given data o. It is rarely available in closed form, and it is what model comparison consumes: the log ratio of two models' evidences is the log Bayes factor, and a reported conclusion rests on that number [Kass and Raftery, 1995, Gelman et al., 2013].

Variational Laplace is the standard fast substitute [Friston et al., 2007, Zeidman et al., 2023]. One finds the best approximate posterior within a parametric family by minimising an upper bound on $-\log p(o)$, then corrects the minimum value by half the log-determinant of the curvature there.

Those two steps are the first two terms of one asymptotic expansion. Write the exponent as an *action*,

$$A(s) \;=\; -\log p(s, o), \qquad p(o) \;=\; \int e^{-A(s)}\, ds, \tag{1}$$

expand A about its minimum s^*, and integrate term by term. The value $A(s^*)$ is the first term. The Gaussian integral of the quadratic term is the second, and equals the log-determinant. The third term has a closed form in the third and fourth derivatives of A at s^*. Writing $\boldsymbol{H}_*$ for the Hessian of A at s^*, $\boldsymbol{G} = \boldsymbol{H}_*^{-1}$ for its inverse, and V_{ijk} and W_{ijkl} for the third- and fourth-derivative tensors there, with indices running over the K coordinates of s and repeated indices summed,

$$F_2 \;=\; \underbrace{\tfrac{1}{8} W_{ijkl} G_{ij} G_{kl}}_{\text{figure-eight}} \;-\; \underbrace{\tfrac{1}{12} V_{ijk} V_{lmn} G_{il} G_{jm} G_{kn}}_{\text{sunset}} \;-\; \underbrace{\tfrac{1}{8} u_i G_{il} u_l}_{\text{spectacles}}, \qquad u_i := V_{ijk} G_{jk}. \tag{2}$$

The matrix $\boldsymbol{G}$ is the approximate posterior covariance that a variational Laplace pipeline already computes. The three names label the shapes of the graphs that generate the three index contractions.

> **TRANSLATION: in one line**
>
> The saddle value $A(s^*)$ is *tree level*; the Occam factor $\frac{1}{2}\log\det \boldsymbol{H}_*$ is the *one-loop* term; $\boldsymbol{G}$ is a *propagator*; V and W are *vertices*; and (2) is the *two-loop* term of the loop expansion of a Euclidean field theory in zero dimensions. All of this is textbook [Negele and Orland, 1988, Zinn-Justin, 2002, Kardar, 2007]. Appendix B is the full correspondence.

Three reasons to compute F_2, in increasing order of weight.

1. *Conceptual.* "Add half a log-determinant" becomes "truncate a series at second order". The error of a truncated asymptotic series is of the order of the first omitted term, and that term is computable.

2. *Practical.* V and W are two more derivatives, by automatic differentiation if convenient, of a function already differentiated twice, at a point already located. In the tests of §15 and §16 the correction removes one to three orders of magnitude of error in the log evidence.

3. *Diagnostic.* $|F_2|$ estimates the error of the number variational Laplace reports. On the Poisson–Gaussian family, across more than an order of magnitude in error size, the ratio of the true one-loop error to $|F_2|$ has median 0.98 and lies in $[0.82, 1.00]$ in 19 of 20 cases (§18). The standard methods reference for this field answers the question "should I trust this?" by recommending a sampling run [Zeidman et al., 2023]; $|F_2|$ answers it for two derivatives.

Two limits belong here rather than later. The series is asymptotic, so adding terms helps and then stops helping (§6.4). And the expansion is about one minimum of A, so it says nothing about a posterior with several widely separated modes (§20).

Part I
The gap

3 Variational Laplace, and what it assumes

3.1 What the evidence is for

A modeller supplies a generative model: a joint density $p(s, o)$ over hidden states s and observations o, factorised as a likelihood times a prior. Two objects are then wanted. The posterior $p(s \mid o) = p(s, o)/p(o)$ is what the data say about the hidden state. The normalising constant

$$p(o) \;=\; \int p(s, o)\, ds \tag{3}$$

is the *model evidence*, also called the marginal likelihood.

The evidence compares two generative models on the same data. The ratio $p(o \mid M_1)/p(o \mid M_2)$ is the Bayes factor and its logarithm is the currency of model selection; Kass and Raftery [1995] is the standard review and Gelman et al. [2013, ch. 7] the standard textbook treatment. In dynamic causal modelling [Friston et al., 2007, Penny, 2012] and in the behavioural fitting of active-inference agents, reported log Bayes factors are routinely a few nats. That is the scale against which everything below should be read.

3.2 The variational free energy

The integral (3) is intractable in most models of interest, and the variational route replaces it by an optimisation [Jordan et al., 1999, Wainwright and Jordan, 2008, Blei et al., 2017]. Pick a tractable family $q_{\boldsymbol{\phi}}(s)$ indexed by parameters $\boldsymbol{\phi} \in \mathbb{R}^N$, for instance all Gaussians, in which case $\boldsymbol{\phi}$ holds a mean and a parameterisation of a covariance. Define the *variational free energy*

$$\mathcal{F}[\boldsymbol{\phi}] \;=\; \mathbb{E}_{q_{\boldsymbol{\phi}}}\big[\log q_{\boldsymbol{\phi}}(s)\big] \;-\; \mathbb{E}_{q_{\boldsymbol{\phi}}}\big[\log p(s, o)\big]. \tag{4}$$

Substituting $p(s, o) = p(s \mid o)\, p(o)$ and taking the constant $\log p(o)$ outside the expectation gives

$$\mathcal{F}[\boldsymbol{\phi}] \;=\; \underbrace{\mathrm{KL}\big[q_{\boldsymbol{\phi}} \,\big\|\, p(\cdot \mid o)\big]}_{\geq 0} \;-\; \log p(o), \qquad \text{so} \qquad \mathcal{F}[\boldsymbol{\phi}] \;\geq\; -\log p(o) \tag{5}$$

for every $\boldsymbol{\phi}$, with equality when $q_{\boldsymbol{\phi}}$ is the exact posterior. Minimising $\mathcal{F}$ therefore moves q towards the posterior and lowers an upper bound on the surprise. The minimiser $\boldsymbol{\phi}^*$ satisfies $\partial \mathcal{F}/\partial \boldsymbol{\phi}|_{\boldsymbol{\phi}^*} = 0$ and is found by gradient descent. Bishop [2006, ch. 10] and MacKay [2003, ch. 33] derive this; MacKay is the clearer source on why the bound is a bound.

3.3 Variational Laplace as practised

A pipeline does not stop at $\mathcal{F}[\boldsymbol{\phi}^*]$. Take q Gaussian with mean μ and covariance Σ, expand $-\log p(s, o)$ to second order about its mode s^*, and optimise (4) over Σ. The optimum is $\Sigma = \boldsymbol{H}_*^{-1}$ with $\boldsymbol{H}_*$ the Hessian of $-\log p(s, o)$ at s^*, and substituting back,

$$-\log p(o) \;\approx\; \underbrace{-\log p(s^*, o)}_{\text{the MAP term}} \;+\; \tfrac{1}{2} \log\det \boldsymbol{H}_* \;-\; \tfrac{K}{2} \log 2\pi, \qquad K = \dim s. \tag{6}$$

This is the classical Laplace approximation [Tierney and Kadane, 1986, MacKay, 1992], and the combination of (4) with (6) is what this literature calls variational Laplace [Beal, 2003, Friston et al., 2007], with Zeidman et al. [2023] as the tutorial reference. The half-log-determinant is

the *Occam factor*: it penalises a posterior that is sharply peaked in many directions, which is how evidence-based comparison penalises complexity [MacKay, 1992, Kass and Raftery, 1995].

Equation (6) assumes that the exponent is quadratic near its minimum. That assumption is exact for a linear-Gaussian model, and approximate inference then coincides with exact inference. Otherwise the third and higher derivatives contribute, and the size of their contribution is not reported. The rest of this document computes it.

One point of vocabulary, since both integrals in this subject receive a Laplace approximation. Everything from here to §11 expands over the *latents* s, where the integral is the evidence itself and every number quoted has a direct reading. A second construction expands instead over the variational parameters ϕ; it is laid out beside this one in §14.1, and nothing before that point depends on it.

3.4 Nats and Bayes factors

A difference of Δ nats between two log evidences is a Bayes factor of e^{Δ}. The conventional reading scale [Kass and Raftery, 1995] runs: below 1 nat, not worth reporting; 1 to 3, positive evidence; 3 to 5, strong; above 5, very strong.

That fixes the scale of the empirical claim. A correction of 10^{-2} nats matters when it is systematic, when it differs between the two models being compared, and when the comparison itself sits at a few nats. Those conditions describe the normal case in the literature this method serves.

4 One worked example, and the gap it leaves

This example carries Parts II and III. It is the smallest problem that is genuinely non-Gaussian, has a ground truth computable to machine precision, and has non-zero third and fourth derivatives.

4.1 The model

An agent runs a single trial and counts how many times an outcome occurs. Its belief about the log-rate $s \in \mathbb{R}$ is a standard normal and the count o is Poisson with rate e^s:

$$p(s) = \tfrac{1}{\sqrt{2\pi}} e^{-s^2/2}, \qquad p(o \mid s) = \frac{e^{-e^s}\, e^{so}}{o!}. \tag{7}$$

The trial returns $o = 0$. The evidence is $p(o) = \int p(o \mid s)\, p(s)\, ds = \int e^{-A(s)}\, ds$, where the *action* $A(s) := -\log p(s, o)$ collects everything in the exponent:

$$A(s) \;=\; \underbrace{e^s - o\,s + \log o!}_{\text{from the likelihood}} \;+\; \underbrace{\tfrac{1}{2}s^2 + \tfrac{1}{2}\log 2\pi}_{\text{from the prior}}. \tag{8}$$

Term by term: e^s is the rate, and penalises large s; $-o\,s$ is the data pulling the other way, a larger count favouring a larger rate; $\log o!$ is constant in s and shifts the evidence without moving the minimum; $\frac{1}{2}s^2$ is the prior pulling s towards zero; $\frac{1}{2}\log 2\pi$ is the prior's normalisation. At $o = 0$ the two data-dependent terms vanish, so everything downstream consists of derivatives of $e^s + \frac{1}{2}s^2$. The integral has no closed form. Adaptive Gauss–Kronrod quadrature gives

$$-\log p(o = 0) \;=\; 0.9630 \text{ nats}. \tag{9}$$

> **TRANSLATION: what has just been written down**
>
> Statistical physics calls $A(s)$ an *action*, $p(o)$ its *partition function*, $-\log p(o)$ its *free energy* and $p(s \mid o)$ the corresponding *Boltzmann measure*. These are not analogies: the integral

$\int e^{-A}$ is a partition function in the ordinary sense of that subject, which is why the methods of Negele and Orland [1988] and Kardar [2007] apply to it unmodified. Mézard and Montanari [2009] develops the same correspondence from the inference side. Appendix B has the rest of it.

4.2 The Laplace approximation

Locate the minimum of A, measure the curvature there, and replace A by a parabola. The stationarity condition is

$$A'(s) = e^s + s - o = 0 \qquad \Longrightarrow \qquad e^{s^*} + s^* = o, \tag{10}$$

transcendental and solved numerically; at $o = 0$, $s^* = -0.5671$. Differentiating again, and writing H_*, V_* and W_* for the second, third and fourth derivatives of A at s^*,

$$H_* = e^{s^*} + 1, \qquad V_* = e^{s^*}, \qquad W_* = e^{s^*}, \qquad G := 1/H_*. \tag{11}$$

The exponential contributes identically to every derivative from the second upward, while the prior contributes only to the second, so V_* and W_* are comparable to H_*. The problem therefore has large vertices in the sense of §7, and the curvature of the prior, the $+1$ in H_*, is what keeps the expansion controlled at small o.

The parabola integrates in closed form, giving

$$-\log p(o) \;\approx\; A(s^*) + \tfrac{1}{2}\log H_* - \tfrac{1}{2}\log 2\pi \;=\; 1.6469 - 0.6943 \;=\; 0.9526. \tag{12}$$

NUMBERS **The worked example at $o = 0$.**

Parts II and III are built from these five numbers (`two_loop_validation.py`, and Table 1 of the paper):

$s^* = -0.5671$	$H_* = 1.5671$	$V_* = W_* = 0.5671$	$G = 1/H_* = 0.6381$

tree $A(s^*) = 1.6469$	+	one loop -0.6943	=	Laplace 0.9526
exact 0.9630		error 0.0104		

4.3 The gap

0.9526 against 0.9630: an error of 0.0104 nats.
Part II accounts for it.

Two remarks before continuing. On this problem 0.0104 nats is small enough to ignore if three decimal places of a log evidence is all that is wanted; the difficulty is that the method does not report the figure, and in other regimes the same quantity is of order one (§17) or exceeds the term it corrects (§6.4). And the Laplace value lies *below* the exact one, although (5) gives a lower bound of exactly that form. That is the subject of §4.4 below.

GOTCHA **The saddle equation is $e^s + s = o$, not $s\,e^s = o$.**

The two are easy to confuse, and the wrong one has a memorable solution: $se^s = 1$ gives the Omega constant 0.5671, which looks plausible. The coincidence survives a quick check, because 0.5671 is also the correct value of e^{s^*} at $o = 0$. At $o = 1$ the correct root is exactly $s^* = 0$, since $e^0 + 0 = 1$. Solve (10) numerically rather than by inspection.

4.4 Does the Kullback–Leibler identity not settle this?

QUESTION **Minimising $\mathcal{F}$ gives a bound and the gap *is* the Kullback–Leibler divergence, by (5). That is an identity. Where does a series enter?**

The identity is exact, and the bound is genuine. The reported number, however, comes from two approximations in sequence:

1. restricting q to a parametric family, whose error is the Kullback–Leibler gap in (5), which no loop order removes;
2. evaluating the resulting expression by expanding the action to second order about the mode, whose error is the truncation error of an asymptotic series.

F_2 is the next term of the second approximation alone.

The numbers above settle the matter arithmetically. If the reported value were the variational minimum it would satisfy $\mathcal{F} \geq -\log p(o)$. It does not: 0.9526 against an exact 0.9630. The bound is violated by the 0.0104 nats this document is about, and the two-loop value of 0.9641 lies above the exact value again. Measured over a parameter sweep, the errors of the exact Gaussian variational optimum are positive throughout, while the one-loop errors change sign. The identity constrains the sign of the error of the first approximation and says nothing about the magnitude of the second.

WHITEBOARD DRILL **Before Part II.**

Write $A(s)$ for this model from the generative model alone, differentiate three times, and confirm $V_* = W_* = e^{s^*}$. Then say in one sentence why $\log o!$ cannot affect s^*, H_*, V_* or W_* while it does affect the evidence.

FURTHER READING: Part I: evidence, variational inference, and the Laplace step

Zeidman et al. [2023] is the direct background: a tutorial on variational Laplace with worked derivations and standalone code. It is also the document §18 argues with, since its prescription for checking the Laplace assumption is to run a sampler.

For variational inference, read Bishop [2006, ch. 10] first for the mean-field derivation, then Blei et al. [2017] as a twenty-page review of modern practice. Murphy [2023, ch. 10] is the current graduate-level account. Wainwright and Jordan [2008] is a long monograph that places variational inference inside exponential-family theory, and is the right reference for why the cumulant structure of §16 and §17 is not a coincidence. Jordan et al. [1999] is the historical paper that fixed the notation.

MacKay [2003] is free online and the best-written book in this list: chapters 27–33 cover Laplace's method, approximating distributions, Occam factors and Monte Carlo, and chapter 28 is the clearest short account of why the log-determinant penalises complexity. Gelman et al. [2013, ch. 7] gives the applied view of evidence and model comparison, and Kass and Raftery [1995] the reading scale for Bayes factors.

For the Laplace approximation as an asymptotic method rather than a formula, Bender and Orszag [1999, ch. 6] is the standard first treatment, Wong [2001, ch. 2] the rigorous one, and Small [2010, ch. 6] the statistician's version with the Bayesian application worked. Barndorff-Nielsen and Cox [1989] and Butler [2007] cover the same ground in the saddlepoint idiom that statistics uses for exactly these expansions.

On the free-energy formulation, Buckley et al. [2017] derives the machinery in ordinary applied notation with the Laplace step written out, and Bogacz [2017] does the same at a gentler slope. Parr et al. [2022] is the book, Smith et al. [2022] a tutorial with code, and Da Costa et al. [2020] treats the discrete case needed for §17.

Part II
The programme, in four steps

5 The programme

	What it produces	What that is
1	the action	The exponent, expanded about its minimum. Everything afterwards consists of derivatives of one function at one point (§6).
2	the Feynman rules	One rule for each term of that expansion: a line for the quadratic term, an n-legged dot for the nth derivative (§7).
3	the diagrams	Every shape that can be built from those pieces at a given order. The list is finite and short, and each shape evaluates to a number (§8).
4	the sum	Their sum, which is the next term of the series (§9).

Textbook presentations of these four steps are Mattuck [1992, ch. 3–4], Negele and Orland [1988, ch. 2], Kardar [2007, ch. 2–3] and Zee [2010, part I]. Veltman [1994] treats the diagrammatic bookkeeping on its own.

6 Step one: expand the action about its minimum

6.1 The split

Write $s = s^* + \eta$, with η the fluctuation about the minimum, and Taylor-expand. There is no linear term, since s^* is stationary:

$$A(s^*+\eta) \;=\; A(s^*) \;+\; \underbrace{\tfrac{1}{2}H_*\eta^2}_{\text{quadratic}} \;+\; \underbrace{\tfrac{1}{3!}V_*\eta^3}_{\text{cubic}} \;+\; \underbrace{\tfrac{1}{4!}W_*\eta^4}_{\text{quartic}} \;+\; \cdots \tag{13}$$

The Laplace approximation keeps the quadratic term and discards the rest. Write X for what it discards:

$$X \;:=\; \tfrac{1}{3!}V_*\eta^3 + \tfrac{1}{4!}W_*\eta^4 + \cdots \tag{14}$$

Restoring X factorises the integral exactly, with no approximation made:

$$\int e^{-A(s)}\,ds \;=\; \underbrace{e^{-A(s^*)}}_{\text{the minimum}} \;\; \underbrace{\sqrt{\tfrac{2\pi}{H_*}}}_{\text{the parabola}} \;\; \underbrace{\big\langle e^{-X}\big\rangle_0}_{\text{the remainder}}\,, \tag{15}$$

where $\langle\cdot\rangle_0$ is the normalised Gaussian average against the parabola alone,

$$\langle f\rangle_0 \;:=\; \frac{\int f(\eta)\, e^{-\frac{1}{2}H_*\eta^2}\,d\eta}{\int e^{-\frac{1}{2}H_*\eta^2}\,d\eta}. \tag{16}$$

The first two factors of (15) are variational Laplace. The third equals 1 exactly when the action is quadratic. Expanding the exponential,

$$\big\langle e^{-X}\big\rangle_0 \;=\; 1 \;-\; \langle X\rangle_0 \;+\; \tfrac{1}{2}\langle X^2\rangle_0 \;-\; \cdots\,, \tag{17}$$

and the remainder of Part II evaluates the first two surviving terms of this series.

6.2 Why the quadratic term is handled separately

QUESTION **Why does η^2 become a line while η^3 and η^4 become dots?**

The quadratic term is the one term that can be integrated in closed form, so it is absorbed into the *measure* and the others are left in the *integrand*.

Being absorbed, the quadratic term is never expanded. What survives of it is its inverse, $G = H_*^{-1} = \langle \eta^2 \rangle_0$. A second moment relates two copies of η, and a line joining two ends is the natural picture for that. The cubic and quartic terms are pulled down from e^{-X} by (17), and each factor η^n arrives with n loose ends awaiting pairing. The distinction is between Gaussian and non-Gaussian, rather than between 2 and $\{3, 4\}$.

The power counting of §6.3 restates this arithmetically: the quadratic term is the only one whose weight is independent of the expansion parameter. If a quartic integral were available in closed form, the quartic term would go into the measure instead.

6.3 Why the fourth derivative is the last one needed

The Taylor series (13) is infinite, so a finite calculation requires a rule for which terms contribute at a given order.

Introduce a bookkeeping parameter $\beta > 0$ multiplying the action, with $\beta = 1$ the case of interest, and rescale the fluctuation as $\eta = \xi/\sqrt{\beta}$ so that the quadratic term loses its β:

$$\beta A(s^* + \xi/\sqrt{\beta}) \;=\; \beta A(s^*) + \tfrac{1}{2} H_* \xi^2 + \underbrace{\frac{V_* \xi^3}{6\sqrt{\beta}}}_{X_3} + \underbrace{\frac{W_* \xi^4}{24\,\beta}}_{X_4} + \cdots \tag{18}$$

Every interaction term now carries an explicit negative power of β: the nth term carries $\beta^{1-n/2}$, so a cubic vertex carries $\beta^{-1/2}$, a quartic vertex β^{-1} and a quintic vertex $\beta^{-3/2}$. At $n = 2$ the exponent is zero, which is what makes the parabola the leading term.

Collect everything of order β^{-1} in $\log \langle e^{-X} \rangle_0$. One quartic vertex qualifies, and so does a pair of cubic vertices. Nothing else does.

Term	β power	Survives?	Reason
$\langle X_3 \rangle_0$	$\beta^{-1/2}$	no	odd moment, vanishes
$\langle X_4 \rangle_0$	β^{-1}	**yes**	the figure-eight
$\frac{1}{2}\langle X_3^2 \rangle_0$	β^{-1}	**yes**	sunset and spectacles
$\langle X_3 X_4 \rangle_0$, $\langle X_5 \rangle_0$	$\beta^{-3/2}$	no	odd, and higher order
$\langle X_4^2 \rangle_0$, $\langle X_6 \rangle_0$, $\langle X_3^4 \rangle_0$	β^{-2}	no	three loops

Two survivors. The second splits into two shapes once indices are restored (§10), so the answer at this order has three diagrams, and this table is the proof that there are no others. The implementation consequence is that the fifth derivative of the action first appears at three loops, so a two-loop scheme never needs a derivative beyond the fourth.

GOTCHA **β is bookkeeping, not a parameter of your model.**

Nothing here requires a choice of β. At $\beta = 1$ the series is still a valid asymptotic expansion, now in the inverse curvature at the saddle: each order carries two or three more powers of $G = H_*^{-1}$ than the last, so the small quantity is the width of the posterior, and hence the amount of data. That is the content of the Bernstein–von Mises theorem [van der Vaart, 2000, ch. 10]. §14 gives β a reading for those who want one.

6.4 The series is asymptotic

The series

$$-\log Z \;\sim\; A(s^*) + F_1 + F_2 + F_3 + \cdots \tag{19}$$

generically has zero radius of convergence. The coefficients F_L grow factorially with the loop order L, because the number of diagrams does. Bender and Wu [1969, 1973] established this for the quartic anharmonic oscillator, and Bender and Orszag [1999, ch. 3] gives the general account.

What an asymptotic series provides instead is a truncation error of the order of the first omitted term, as the expansion parameter becomes small. There is therefore an optimal truncation order, beyond which further terms move the estimate away from the answer, and an accuracy floor that no additional work improves. At one or two loops on a well-behaved posterior the series behaves like a convergent one.

The useful output is an answer to "is the next term small?", which is §18. The claim that the truncation error is the size of the first omitted term is measured in §9.2, once that term has been computed.

> **QUESTION** **Does the series converge?**
>
> No, and no claim here depends on its converging. Partial sums approximate the answer to within the next omitted order, so additional terms help, then stop helping, then hurt. Three consequences. A two-loop number quoted without an estimate of the expansion parameter is incomplete. Adding orders is not a strategy for a hard problem. And when the expansion does break down, the breakdown is visible in the numbers already computed. Boyd [1999] is a readable review of what divergent series are good for.

WHITEBOARD DRILL **After step one.**

Derive (15) from (13) in two lines, and say where the normalisation of $\langle\cdot\rangle_0$ comes from. Then reconstruct the order-counting table from the single fact that the nth term of (18) carries $\beta^{1-n/2}$, and use it to answer: at what loop order does the sixth derivative of the action first appear?

7 Step two: the Feynman rules

A Feynman rule is a piece of notation standing for a number. There is one rule for each term of (13), so for this action there are three.

	a **propagator** from the $\frac{1}{2}H_*\eta^2$ term	$G = 1/H_* = 0.638$
	a **three-leg vertex** from the $\frac{1}{6}V_*\eta^3$ term	$V_* = 0.567$
	a **four-leg vertex** from the $\frac{1}{24}W_*\eta^4$ term	$W_* = 0.567$

A line carries the variance of the parabola. A dot with n legs carries the nth derivative of the action at the saddle, with one leg for each index. The $1/n!$ in front of each Taylor term belongs to the counting at step three rather than to the rule, and §8 does that counting by enumeration. The standard compact statements of these rules are Peskin and Schroeder [1995, §4.4] and Negele and Orland [1988, ch. 2]; the longest and gentlest is Mattuck [1992, ch. 3–4].

7.1 Wick's theorem

The content of "a line carries G" is a theorem about Gaussian moments.

Lemma 1 (Wick, Isserlis)**.** *Let $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{G})$ in $\mathbb{R}^K$. Then for any indices $i_1, \dots, i_{2m}$,*

$$\big\langle \xi_{i_1} \xi_{i_2} \cdots \xi_{i_{2m}} \big\rangle_0 \;=\; \sum_{P \in \mathcal{P}_{2m}} \; \prod_{(a,b) \in P} G_{i_a i_b}, \tag{20}$$

where $\mathcal{P}_{2m}$ is the set of perfect matchings, or pairings*, of the $2m$ slots, of which there are $(2m-1)!! = 1, 3, 15, 105, \dots$ for $2m = 2, 4, 6, 8$. Odd moments vanish.*

Proof. Completing the square in the Gaussian integral gives the moment generating function $M(\boldsymbol{J}) = \langle e^{\boldsymbol{J}^\top \boldsymbol{\xi}} \rangle_0 = \exp\!\big(\tfrac{1}{2} \boldsymbol{J}^\top \boldsymbol{G} \boldsymbol{J}\big)$, so moments are derivatives at $\boldsymbol{J} = \mathbf{0}$. Each derivative ∂_{J_i} either acts on the exponent, producing a factor $(\boldsymbol{G}\boldsymbol{J})_i$ which must be differentiated again to survive at $\boldsymbol{J} = \mathbf{0}$, or acts on a previously produced factor, consuming it and yielding G_{ij}. Surviving terms therefore pair up all the derivatives, so their number must be even, and each pairing contributes one product of G's. Counting matchings of $2m$ objects: the first slot has $2m - 1$ possible partners, then recurse, giving $(2m-1)!!$. □

The identity is due to Isserlis [1918] in the statistics literature and to Wick in the physics one; Negele and Orland [1988, ch. 2] and Mattuck [1992, ch. 3] give the diagrammatic reading used below. McCullagh [1987, ch. 2] is the systematic statistical treatment of moment and cumulant identities in index notation, and is the closest thing in statistics to the bookkeeping of this primer.

The two cases needed here, written out:

$$\langle \xi_i \xi_j \rangle_0 = G_{ij}, \tag{21}$$

$$\langle \xi_i \xi_j \xi_k \xi_l \rangle_0 = G_{ij} G_{kl} + G_{ik} G_{jl} + G_{il} G_{jk} \qquad \text{(3 terms)}, \tag{22}$$

$$\langle \xi_i \xi_j \xi_k \xi_l \xi_m \xi_n \rangle_0 = G_{ij} G_{kl} G_{mn} + \cdots \qquad \text{(15 terms)}, \tag{23}$$

the fifteen enumerated in §8.4. In one dimension every G is the same number and (20) collapses to

$$\langle \eta^{2m} \rangle_0 \;=\; (2m-1)!!\; G^m, \qquad \text{so} \qquad \langle \eta^4 \rangle_0 = 3G^2, \quad \langle \eta^6 \rangle_0 = 15 G^3. \tag{24}$$

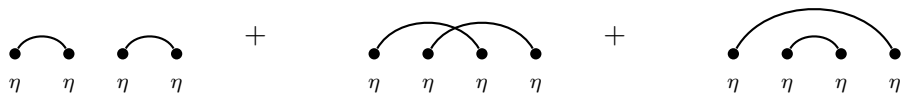


Figure 1: Wick's theorem at fourth order. A Gaussian average of a product equals the sum over every way of pairing the factors, each pair carrying one $G = \langle \eta^2 \rangle_0$. A line in a diagram is one of these pairings, which is what makes a diagram a calculation. Four copies of η pair in three ways; six pair in fifteen.

TRANSLATION: a result you may already use

Wick's theorem is Isserlis' theorem, taught in multivariate statistics as the moment identity for Gaussian vectors. Anyone who has written $\mathbb{E}[x_i x_j x_k x_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$ for a zero-mean Gaussian has used it. The two literatures kept separate names.

NUMBERS **Wick's theorem, checked to machine precision. Exercise 1.**

The script verifies (20) twice, by exact quadrature rather than sampling. In one dimension it compares $(2m-1)!!\, G^m$ against numerical quadrature for $2m = 2, 4, 6, 8$ with $G = 2$, giving 2, 12, 120, 1680 to relative error $\le 2 \cdot 10^{-16}$. In three dimensions it evaluates all 81

fourth moments $\langle \xi_i \xi_j \xi_k \xi_l \rangle_0$ by tensor-product Gauss–Hermite quadrature, which is exact for polynomials, and compares with (22): worst absolute error $1.6 \cdot 10^{-14}$. It also checks three sixth moments against the full 15-pairing sum, worst error $3.2 \cdot 10^{-14}$.

GOTCHA **Wick's theorem pairs *slots*, not distinct indices.**

In $\langle \xi_1 \xi_1 \xi_2 \xi_2 \rangle_0$ the four slots hold indices $1, 1, 2, 2$, and there are still exactly three pairings; they evaluate to $G_{11}G_{22} + G_{12}G_{12} + G_{12}G_{12}$, so two of them coincide numerically. Coincidences of this kind are the source of every factor-of-k error in this subject, including the one §12 is about. Enumerate pairings of slots first, then substitute indices.

7.2 Why the expansion is of $\log Z$

Expanding e^{-X} and averaging gives a series for Z. The quantity wanted is $\log Z$. The relation between them is the *cumulant expansion*: for a random variable X,

$$\log \big\langle e^{-X} \big\rangle_0 \;=\; -\langle X \rangle_0 \;+\; \tfrac{1}{2}\big[\langle X^2 \rangle_0 - \langle X \rangle_0^2\big] \;-\; \tfrac{1}{6}\big[\text{third cumulant}\big] + \cdots \tag{25}$$

Lemma 2 (Cumulant expansion)**.** *For a random variable X with all moments finite,*

$$\log \big\langle e^{-X} \big\rangle_0 = \sum_{n \geq 1} \frac{(-1)^n}{n!}\, \kappa_n(X),$$

with $\kappa_1 = \langle X \rangle_0$ and $\kappa_2 = \langle X^2 \rangle_0 - \langle X \rangle_0^2$, and so on.

Proof. Both sides are formal power series in X; expand $\log\big(1 + \sum_{n \geq 1} \frac{(-1)^n}{n!} \langle X^n \rangle_0\big)$ and collect. The resulting coefficients are the cumulants by definition. □

McCullagh [1987, ch. 2] treats cumulants and their index notation at length, and Wainwright and Jordan [2008, ch. 3] gives the exponential-family version used in §16.

The interpretation is the *linked-cluster theorem*. A term in $\langle X^n \rangle_0$ corresponds to a diagram with n vertices, which may or may not be connected. The subtractions built into κ_n are the products of lower-order diagrams, which are the disconnected ones. Hence

$$\log Z \;=\; \log Z_0 \;+\; \sum_{\substack{\text{connected} \\ \text{vacuum diagrams}}} (\text{value}), \tag{26}$$

proved diagrammatically in Mattuck [1992, ch. 4] and Negele and Orland [1988, ch. 2].

TRANSLATION: an instance

At the order after the one computed here, $\frac{1}{2}\langle X^2 \rangle_0$ contains $\frac{1}{2}\langle X_4 \rangle_0^2$: two separate figure-eights side by side, a disconnected diagram. The cumulant subtraction removes it exactly. Working with Z instead of $\log Z$ generates such terms and then cancels them, which is why the evidence expansion is organised as an expansion of the log evidence.

TRANSLATION: a simplification at this order

In $\frac{1}{2}[\langle X_3^2 \rangle_0 - \langle X_3 \rangle_0^2]$ the subtraction can be dropped, because $\langle X_3 \rangle_0 = 0$ by parity. That holds for a cubic vertex at this order; in general the subtraction is required. The graph-theoretic statement, keep connected diagrams, and the statistical statement, use cumulants rather than moments, are the same statement.

7.3 What the rules build

The rules are pieces. Combining them gives *diagrams*, and each diagram evaluates to a number. Which diagrams are required depends on the quantity being computed.

Suppose the quantity wanted is $\langle \eta^k \rangle$ rather than the normalisation. The expansion then has k factors of η outside the exponential, and after Wick pairing k ends of lines remain unattached. Those are *external legs*, and their number is fixed by the quantity being computed rather than by the order of the calculation.

- **Two legs** gives the variance, $\langle \eta^2 \rangle$.
- **One leg** gives the shift of the peak, $\langle \eta \rangle$: the difference between the mode and the posterior mean.
- **No legs** gives the evidence, which averages no η at all, so every line closes on itself.

Diagrams with no external legs are *vacuum diagrams*, and they are the ones contributing to $\log Z$. The evidence is the subject of this primer, so every diagram from §8 onward is of that kind.

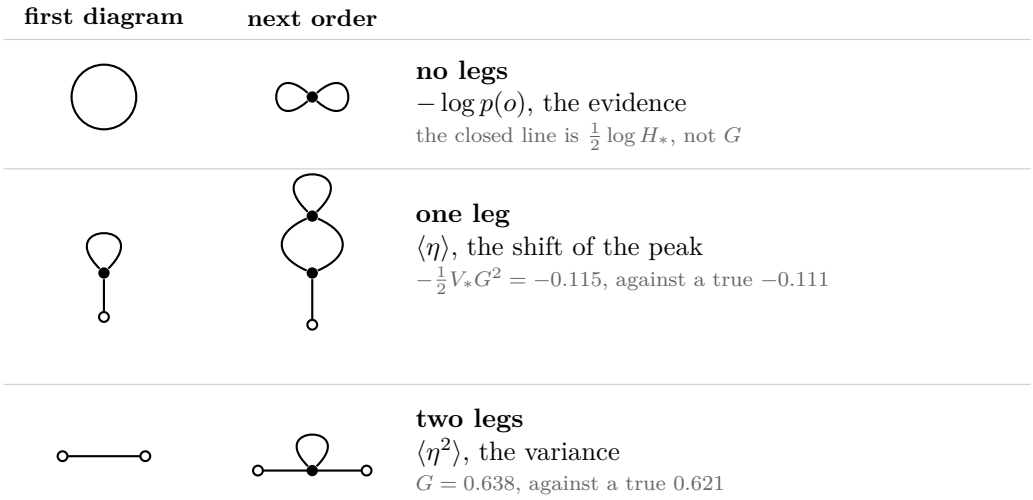


Figure 2: External legs count the quantity being computed. Open circles are external points. Each row is a different quantity with its own series; the left column shows its leading diagram and the right column the next order. Numbers are the worked example at $o = 0$.

GOTCHA **The one-loop ring and the propagator are the same line in different topologies.**

Leaving the two ends of a line free gives $\langle \eta^2 \rangle_0 = G$: two external legs, no closed circuit, tree level. Joining the two ends to each other gives the ring: no legs, one closed circuit, one loop, with value $\frac{1}{2}\log H_*$.

The ring arises from the Gaussian normalisation $\sqrt{2\pi/H_*}$ rather than from a Wick contraction, and closing a propagator on itself is the standard picture for a log-determinant. Wick contractions supply G per pair; the determinant comes from the measure. Both are leading order in their own column, which is why they occupy the same position in Figure 2.

7.4 The subscript on the average

QUESTION **If $\langle \eta^2 \rangle = G$, what is the two-leg diagram with a quartic vertex in Figure 2, which is also the variance?**

The two averages are taken under different measures, and the subscript distinguishes them. $\langle \cdot \rangle_0$ is the average under the parabola alone, defined in (16), where there is no interaction and $\langle \eta^2 \rangle_0 = 1/H_* = G$ exactly. That is the propagator: an input to the calculation.

$\langle \cdot \rangle$ with no subscript is the expectation under the exact posterior, that is, under $e^{-\frac{1}{2}H_*\eta^2 - X}$. Expanding e^{-X} gives $\langle \eta^2 \rangle = G + (\text{a correction}) + \cdots$, the correction being the quartic diagram in the next-order column of Figure 2. On the worked example at $o = 0$, $G = 0.638$ against a true posterior variance of 0.621, a difference of 2.7%.

Every average in the expansion carries the subscript for this reason. The method computes exact quantities as free averages of the interaction.

WHITEBOARD DRILL **After step two.**

Do the multivariate Gaussian integral $\int e^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{H}\boldsymbol{x}} d^K x$ by diagonalising, and identify the result. Write $\langle \xi_i \xi_j \xi_k \xi_l \rangle_0$ from memory and say how many terms $\langle \xi^6 \rangle_0$ has. Then say in one sentence why the expansion is of $\log Z$, and in one more why the ring carries $\frac{1}{2}\log H_*$ rather than G.

8 Step three: the diagrams and their symmetry factors

8.1 The first diagram

The ring, a single propagator closed on itself with no vertices, is one loop, and its value is $\frac{1}{2}\log H_*$. That is the Occam factor of (6). Variational Laplace is the first diagram of the series, drawn; nothing has been added yet.

8.2 The three shapes at the next order

At the next order the table of §6.3 admits one quartic vertex or two cubic vertices. Up to topology that gives three connected vacuum shapes.



one loop	**figure-eight**	**sunset**	**spectacles**
variational Laplace	$\mathcal{S} = 8$	$\mathcal{S} = 12$	$\mathcal{S} = 8$
$\frac{1}{2}\log H_*$	$+\frac{1}{8}W_* G^2$	$-\frac{1}{12}V_*^2 G^3$	$-\frac{1}{8}V_*^2 G^3$

Figure 3: The complete list at this order, in the one-dimensional example. The ring is the term variational Laplace already computes. Each of the three new shapes has two independent cycles, which is what two loops means, and each carries a symmetry factor $\mathcal{S}$ dividing its value. In one dimension the sunset and the spectacles are the same scalar $V_*^2 G^3$ and combine to $\frac{5}{24}$; in more than one dimension they are different contractions (§12).

A *loop* is a closed circuit in the graph, and each of the three new shapes has two. §8.5 shows that the loop count and the order in the series are the same integer.

8.3 One symmetry factor, counted explicitly

The sunset has two three-leg vertices joined by three propagators, so its value is built from $V_*^2 G^3$ divided by its *symmetry factor* $\mathcal{S}$:

$$\tfrac{1}{12}\, V_*^2\, G^3 \;=\; 0.0070, \qquad V_* = 0.567, \quad G = 0.638. \tag{27}$$

Here $\mathcal{S} = 12$ is 3! for permuting the three lines between the vertices, times 2 for exchanging the two vertices. The figure-eight has $\mathcal{S} = 8$ and the spectacles $\mathcal{S} = 8$.

TRANSLATION: what a symmetry factor is

$\mathcal{S} = |\mathrm{Aut}|$, the order of the automorphism group of the graph: the number of relabellings of its lines and vertices that leave the picture unchanged. Equivalently, $1/\mathcal{S}$ is the number of Wick pairings producing that contraction divided by the vertex factorials from (13), and the second form is the one to compute with. Mattuck [1992, ch. 4] counts them by enumeration and Peskin and Schroeder [1995, §4.4] states the general rules.

8.4 The fifteen pairings, enumerated

The two cubic coefficients come from the following count, done by enumeration.

The sixth moment (23) is required, with the first three slots belonging to one vertex, call it A, and the last three to the other, call it B. Label A's legs a_1, a_2, a_3 and B's legs b_1, b_2, b_3. There are fifteen pairings.

A cubic vertex has an odd number of legs, so if p of A's legs pair internally then $3-2p$ must cross to B. The number of crossing edges is therefore 3 or 1, and there is no third topology, hence exactly two cubic diagrams.

Sunset: 3 crossing edges	**Spectacles: 1 crossing edge**
$(a_1b_1)(a_2b_2)(a_3b_3)$	$(a_1a_2)(a_3b_1)(b_2b_3)$
$(a_1b_1)(a_2b_3)(a_3b_2)$	$(a_1a_2)(a_3b_2)(b_1b_3)$
$(a_1b_2)(a_2b_1)(a_3b_3)$	$(a_1a_2)(a_3b_3)(b_1b_2)$
$(a_1b_2)(a_2b_3)(a_3b_1)$	$(a_1a_3)(a_2b_1)(b_2b_3)$
$(a_1b_3)(a_2b_1)(a_3b_2)$	$(a_1a_3)(a_2b_2)(b_1b_3)$
$(a_1b_3)(a_2b_2)(a_3b_1)$	$(a_1a_3)(a_2b_3)(b_1b_2)$
	$(a_1b_1)(a_2a_3)(b_2b_3)$
	$(a_1b_2)(a_2a_3)(b_1b_3)$
	$(a_1b_3)(a_2a_3)(b_1b_2)$
$6 = 3!$ pairings	$9 = 3 \times 3$ pairings (total $15 = 5!!$)

The sunset's six are the 3! bijections between A's legs and B's. The spectacles' nine are three choices of internal pair in A times three in B, with the leftover legs forced to cross; a self-pairing in one vertex forces one in the other, which is why no intermediate case exists. Multiplying by the common prefactor $\frac{1}{2!}\left(\frac{1}{3!}\right)^2 = \frac{1}{72}$ turns the counts into coefficients:

$$\frac{6}{72} = \frac{1}{12} \;\text{(sunset)}, \qquad \frac{9}{72} = \frac{1}{8} \;\text{(spectacles)}, \qquad \frac{1}{12} + \frac{1}{8} = \frac{5}{24} \;\text{(in one dimension)}. \tag{28}$$

QUESTION **Wick's theorem gave fifteen pairings. Where does six come from?**

Fifteen is the number of pairings of six slots. In $\frac{1}{2}\langle X_3^2\rangle_0$ the six slots are not interchangeable: three belong to one vertex and three to the other, and the number of edges crossing between the vertices is a property of a pairing that can be read off the picture. Six pairings cross three times, nine cross once. In one dimension all fifteen evaluate to G^3 and the distinc-

tion is invisible. In more than one dimension the two groups give different tensor contractions, which is §12.

NUMBERS **The enumeration, by machine. Exercise 2.**

`primer_examples.py` generates all perfect matchings of the six legs, classifies them by crossing number, and prints them: 15 matchings in total, 6 with crossing number 3, 9 with crossing number 1, and 0 with any other crossing number. It then prints $6/72 = 0.0833\ldots = 1/12$ and $9/72 = 0.125 = 1/8$. Enumeration settles any doubt about a symmetry factor in ten lines of code.

8.5 Loop counting

Consider a connected vacuum diagram with V_3 cubic vertices, V_4 quartic vertices, E edges and L independent cycles. Two counting identities hold. Every leg is one end of an edge, so $3V_3 + 4V_4 = 2E$; and for a connected graph, $L = E - (V_3 + V_4) + 1$.

From (18) each cubic vertex carries $\beta^{-1/2}$ and each quartic vertex β^{-1}, so the diagram's weight is $\beta^{-(V_3/2+V_4)}$. Eliminating E between the two identities gives

$$\tfrac{1}{2}V_3 + V_4 \;=\; L - 1, \qquad \text{so the weight is} \quad \beta^{-(L-1)}. \tag{29}$$

Loop number and order in the series are the same integer. At $L = 2$ the only solutions of $\frac{1}{2}V_3 + V_4 = 1$ are $(V_3, V_4) = (2, 0)$, giving the sunset and the spectacles, and $(0, 1)$, giving the figure-eight, so Figure 3 is complete. The argument is two lines rather than an exhaustive search, which is the main practical reason to draw the diagrams at all.

At three loops, $\frac{1}{2}V_3 + V_4 = 2$ admits $(4, 0)$, $(2, 1)$ and $(0, 2)$, with five- and six-leg vertices entering as well; the number of connected topologies grows quickly and the fifth and sixth derivatives appear. Zinn-Justin [2002, ch. 9] gives the generating-functional machinery for that case, which is the right tool there; hand enumeration is not.

GOTCHA **Loop order is not the order of accuracy for an expectation.**

For $\log Z$, L loops gives order $\beta^{-(L-1)}$. For an expectation with external legs, such as the mean shift of §13, the counting is shifted, because the diagram is not a vacuum diagram. Redo the count rather than transferring the rule.

WHITEBOARD DRILL **After step three.**

Draw the three two-loop shapes. For each, count edges and vertices and check $L = E - V_{\text{tot}} + 1 = 2$. Then derive the coefficient of each by stating how many of the 15, or of the 3, pairings produce it and what the vertex prefactor is. Then say in one sentence why there is no fourth diagram at this order.

9 Step four: sum the diagrams

9.1 The arithmetic

Three shapes, two distinct values in one dimension, one sum, using $\langle \eta^4 \rangle_0 = 3G^2$ for the quartic vertex and $\langle \eta^6 \rangle_0 = 15G^3$ for the pair of cubic ones.

Diagram	Source	Value at $o=0$
$+\frac{1}{8}W_*G^2$	figure-eight, from $\langle\eta^4\rangle_0 = 3G^2$	$+0.0289$
$-\frac{5}{24}V_*^2G^3$	sunset and spectacles, from $\langle\eta^6\rangle_0 = 15G^3$	-0.0174
F_2	two loops	$+0.0115$

Laplace alone	Laplace and two loops	exact
0.9526	0.9641	0.9630
error 0.0104	error 0.0011	by quadrature

The error falls by a factor of ten, from two more derivatives of a function already differentiated twice, evaluated at a point already located. Those two terms are the whole of the two-loop correction, and the accounting promised in §4 is closed.

GOTCHA **F_2 is a difference of larger numbers, so the coefficients must be right.**

Here $+0.0289$ and -0.0174 are each two to three times larger than their sum. The quartic and the cubic diagrams enter with opposite signs and comparable magnitudes, so a wrong symmetry factor is not a small perturbation. At $o=1$ (§9.3) using $\frac{1}{12}$ where $\frac{5}{24}$ belongs makes F_2 four times too large; in the example of §9.2 it changes the sign. A cancellation at this order also carries no information about the next order, so a defensible error estimate reports the individual diagrams as well as their sum.

9.2 The same calculation without diagrams

The whole calculation in one dimension takes half a page and no pictures, which locates what the pictures are for. Take the general one-dimensional action

$$A(x) \;=\; \tfrac{1}{2}Hx^2 + \tfrac{1}{6}Vx^3 + \tfrac{1}{24}Wx^4, \qquad x\in\mathbb{R}, \tag{30}$$

so that the minimum sits at $x^*=0$ with $A(x^*)=0$ and $G=1/H$. Write

$$\mathcal{E}(\beta) \;:=\; -\tfrac{1}{\beta}\log\int e^{-\beta A(x)}\,dx.$$

Rescale $x=\xi/\sqrt{\beta}$ as in (18) and write $Z = Z_0\langle e^{-X_3-X_4}\rangle_0$ with $Z_0=\beta^{-1/2}\sqrt{2\pi/H}$. Apply the cumulant expansion (25) and keep everything of order β^{-1}, which by the table of §6.3 is two terms:

$$-\langle X_4\rangle_0 = -\frac{W}{24\beta}\langle\xi^4\rangle_0 = -\frac{W}{24\beta}\cdot 3G^2 = -\frac{1}{\beta}\cdot\frac{WG^2}{8}, \tag{31}$$

$$+\tfrac{1}{2}\langle X_3^2\rangle_0 = \frac{1}{2}\cdot\frac{V^2}{36\beta}\langle\xi^6\rangle_0 = \frac{V^2}{72\beta}\cdot 15G^3 = +\frac{1}{\beta}\cdot\frac{5V^2G^3}{24}. \tag{32}$$

Therefore

$$\boxed{\begin{aligned} F_1 &= \tfrac{1}{2}\log H - \tfrac{1}{2}\log(2\pi/\beta), \\ F_2 &= \tfrac{1}{8}WG^2 \;-\; \tfrac{5}{24}V^2G^3. \end{aligned}} \tag{33, 34}$$

Two Gaussian moments and one cumulant identity suffice, and nothing was drawn.

TRANSLATION: what the pictures are for

In one dimension they are unnecessary. They earn their place in three ways, none of them visible yet.

Indices. In more than one dimension the picture identifies which index contraction a term is, and §12 is an entire section about the error that follows from losing track.

Growth. At order k the number of contractions grows like K^k while the number of shapes

does not, so the bookkeeping is uniform in the number of parameters [Negele and Orland, 1988, ch. 2].

Transfer. A family of shapes can be resummed; a list of contractions cannot. Large-K counting, Borel analysis and renormalisation are phrased in shapes, and Zinn-Justin [2002, ch. 9] and Aniceto et al. [2019] are inaccessible without them.

GOTCHA **Keep $W > 0$ in (30).**

With $W = 0$ and $V \neq 0$ the integral diverges, since $e^{-\beta A}$ grows without bound at one end. The quartic term is what makes Z exist. This is the zero-dimensional form of the fact that the loop expansion is an expansion of a well-defined integral; a perturbation series in V alone is a series for an object that does not exist, which is one route to understanding why such series are asymptotic [Bender and Orszag, 1999, ch. 3].

NUMBERS **Zero-dimensional field theory. Exercises 3 and 4.**

The script fixes $H = 1$, so $G = 1$, and computes $\mathcal{E}(\beta)$ by quadrature in the rescaled variable ξ, which keeps the integrand of order one in width at every β and avoids roundoff.

Gaussian check. With $V = W = 0$ and $H = 1.3$, one loop is exact: at $\beta = 1, 5, 20$ the one-loop prediction and the exact value agree to $2 \cdot 10^{-16}$, $4 \cdot 10^{-17}$, $1 \cdot 10^{-17}$. This is the statement that variational Laplace is exact for linear-Gaussian models, measured.

Pure quartic, $W = 1$, $V = 0$, so $F_2 = 1/8$ exactly.

β	1-loop	2-loop	exact	err 1L	err 2L	$\mathrm{err}_{1L} \times \beta^2$
1	−0.918939	−0.793939	−0.834381	$8.46 \cdot 10^{-2}$	$4.04 \cdot 10^{-2}$	0.0846
2	−0.286182	−0.254932	−0.261549	$2.46 \cdot 10^{-2}$	$6.62 \cdot 10^{-3}$	0.0985
5	−0.022844	−0.017844	−0.018377	$4.47 \cdot 10^{-3}$	$5.34 \cdot 10^{-4}$	0.1117
10	0.023235	0.024485	0.024412	$1.18 \cdot 10^{-3}$	$7.37 \cdot 10^{-5}$	0.1176
20	0.028946	0.029259	0.029249	$3.03 \cdot 10^{-4}$	$9.77 \cdot 10^{-6}$	0.1211
50	0.020741	0.020791	0.020791	$4.94 \cdot 10^{-5}$	$6.49 \cdot 10^{-7}$	0.1234
100	0.013836	0.013849	0.013849	$1.24 \cdot 10^{-5}$	$8.22 \cdot 10^{-8}$	0.1242

The last column is the check promised in §6.4. If the one-loop truncation error is the two-loop term, then $|\mathrm{error}_{\text{1-loop}}| \times \beta^2$ must converge to $|F_2|$, and it converges to 0.125. The error of the method in standard use, multiplied by the square of the expansion parameter, converges to the coefficient computed in (34), and the check requires no diagrams.

With a cubic term, $V = 0.8$, $W = 0.5$, so $F_2 = \frac{1}{8}(0.5) - \frac{5}{24}(0.64) = -0.070833$. Using the coefficient $\frac{1}{12}$ in place of $\frac{5}{24}$ gives $F_2 = +0.009167$, a different sign.

9.3 Worked by hand, in exact fractions

At $o = 1$ the saddle equation $e^{s^*} + s^* = 1$ has the root $s^* = 0$, so every number is exact:

$$e^{s^*} = 1, \qquad H_* = 2, \qquad G = \tfrac{1}{2}, \qquad V_* = W_* = 1,$$

$$
\begin{aligned}
A(0) &= 1 + \tfrac{1}{2}\log 2\pi && = 1.918939,\\
\tfrac{1}{2}\log H_* - \tfrac{1}{2}\log 2\pi &= \tfrac{1}{2}\log 2 - \tfrac{1}{2}\log 2\pi && = -0.572365,\\
\text{one loop, i.e. Laplace} &= A(0) + \tfrac{1}{2}\log\tfrac{H_*}{2\pi} && = 1.346574,\\
\text{figure-eight} &= \tfrac{1}{8}\cdot 1\cdot\left(\tfrac{1}{2}\right)^2 = \tfrac{1}{32} && = +0.03125000,\\
\text{sunset and spectacles} &= -\tfrac{5}{24}\cdot 1\cdot\left(\tfrac{1}{2}\right)^3 = -\tfrac{5}{192} && = -0.02604167,\\
F_2 &= \tfrac{1}{32} - \tfrac{5}{192} = \tfrac{1}{192} && = +0.00520833,\\
\text{two loops} &= 1.346574 + \tfrac{1}{192} && = 1.351782,
\end{aligned}
\tag{35}
$$

against the exact $-\log p(1) = 1.351483$ from quadrature. The one-loop error is 0.004909 nats and the two-loop error is 0.000299, a reduction by a factor of sixteen. As at $o = 0$, $F_2 = 1/192$ is the difference of $1/32$ and $5/192$, two numbers about six times larger than their difference.

WHITEBOARD DRILL **The central drill.**

From (30), derive (33) and (34). Two facts may be looked up:

$$
\int e^{-Hx^2/2}\,dx = \sqrt{2\pi/H}, \qquad \langle x^{2m}\rangle_0 = (2m-1)!!\,G^m.
$$

Reconstruct the rest: the rescaling, which cumulants survive, and the arithmetic $3/24 = 1/8$ and $15/72 = 5/24$. Then say which of the two terms in F_2 is the figure-eight and what its sign is, and work §9.3 without a calculator.

FURTHER READING: Part II: diagrams, Gaussian moments, and asymptotic series

Mattuck [1992] is the book to read if only one is read. It is a cheap Dover paperback whose stated purpose is teaching Feynman diagrams to people who do not know them; it treats non-relativistic many-body theory, so no relativity intervenes, and chapters 3–4 contain everything used here, with symmetry factors counted by explicit enumeration as in §8.4.

For the same material in a statistical-mechanics idiom, with no factors of i and no Wick rotation, Kardar [2007, ch. 2–3] is the closest to what this primer needs, and Negele and Orland [1988, ch. 2] does the vacuum-diagram case in the Euclidean setting. Zee [2010, part I, §§1–7] is the friendliest full field-theory text and Srednicki [2007], free on the author's website, the most systematic. Veltman [1994] is a short book on the diagrammatic rules alone. Altland and Simons [2010, ch. 1–2] derives the functional integral and the perturbation series from scratch for a condensed-matter audience, and is useful if the physics vocabulary is new.

On the statistics side, McCullagh [1987, ch. 2–3] is the systematic treatment of moments, cumulants and index notation, and is the nearest statistical relative of the bookkeeping here. Wainwright and Jordan [2008] provides the exponential-family frame in which the vertices of §16 and §17 are cumulants by construction.

On asymptotic series, Bender and Orszag [1999, ch. 3] is the standard treatment and covers optimal truncation. Boyd [1999] is a 98-page review of what divergent series are good for and is the right calibration before anyone proposes resumming one. Bender and Wu [1969, 1973] are the original anharmonic-oscillator analyses behind the factorial growth, and Aniceto et al. [2019] is the modern resurgence machinery, well beyond what is used here. Mézard and Montanari [2009] is the bridge book between statistical physics and inference, and its chapter 2 is the best short account of what a temperature means when the system is an inference problem.

Part III
The general case

10 From one latent to many

Only the number of latents changes. Every object of the worked example has a counterpart, the derivatives become tensors with one index per leg, and a propagator joining two legs becomes a sum over the index they share.

Worked example	**General case**	
one number s	$s \in \mathbb{R}^K$	the latent states
$A(s) = -\log p(s, o)$	the same, in K variables	still the action
$H_* = A''(s^*)$	$[\boldsymbol{H}_*]_{ij} = \partial^2 A/\partial s_i \partial s_j$	a matrix
V_*, W_*	V_{ijk}, W_{ijkl}	tensors, one index per leg
$G = 1/H_*$	$\boldsymbol{G} = \boldsymbol{H}_*^{-1}$	the posterior covariance

10.1 Setup

Let $s = s^* + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \in \mathbb{R}^K$ and Taylor-expand, with repeated indices summed over $1, \dots, K$ throughout:

$$A(s^* + \boldsymbol{\eta}) \;=\; A(s^*) + \tfrac{1}{2}\boldsymbol{\eta}^\top \boldsymbol{H}_* \boldsymbol{\eta} + \tfrac{1}{3!} V_{ijk}\eta_i\eta_j\eta_k + \tfrac{1}{4!} W_{ijkl}\eta_i\eta_j\eta_k\eta_l + \mathcal{O}(\eta^5), \tag{36}$$

where

$$[\boldsymbol{H}_*]_{ij} = \left.\frac{\partial^2 A}{\partial s_i \partial s_j}\right|_{s^*}, \qquad V_{ijk} = \left.\frac{\partial^3 A}{\partial s_i \partial s_j \partial s_k}\right|_{s^*}, \qquad W_{ijkl} = \left.\frac{\partial^4 A}{\partial s_i \partial s_j \partial s_k \partial s_l}\right|_{s^*}. \tag{37}$$

The linear term vanishes because s^* is stationary. All three tensors are fully symmetric in their indices, since partial derivatives commute. That symmetry is used repeatedly below and is worth checking numerically in any implementation: if it fails, the derivatives are wrong.

Rescaling $\boldsymbol{\eta} = \boldsymbol{\xi}/\sqrt{\beta}$ as in (18) gives

$$Z(\beta) = e^{-\beta A(s^*)}\beta^{-K/2} \int e^{-\frac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{H}_* \boldsymbol{\xi} - X_3 - X_4 - \cdots}\, d^K\xi, \qquad \begin{aligned} X_3 &= \tfrac{1}{6\sqrt{\beta}} V_{ijk}\xi_i\xi_j\xi_k, \\ X_4 &= \tfrac{1}{24\beta} W_{ijkl}\xi_i\xi_j\xi_k\xi_l. \end{aligned} \tag{38}$$

Dropping the vertices and using $\int e^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{H}\boldsymbol{x}} d^K x = (2\pi)^{K/2} (\det \boldsymbol{H})^{-1/2}$ gives the one-loop result, which is (6).

10.2 The figure-eight

By (22), and because W is symmetric, all three pairings of ξ^4 give the same contraction:

$$\langle X_4 \rangle_0 = \frac{W_{ijkl}}{24\beta}\big(G_{ij}G_{kl} + G_{ik}G_{jl} + G_{il}G_{jk}\big) = \frac{3}{24\beta} W_{ijkl}G_{ij}G_{kl} = \frac{1}{\beta} \cdot \frac{W_{ijkl}G_{ij}G_{kl}}{8}. \tag{39}$$

One quartic vertex, its four legs paired into two self-loops: two circles meeting at a point, which is the name. Edges $E = 2$, vertices $V_{\text{tot}} = 1$, so $L = 2 - 1 + 1 = 2$ by (29). The symmetry of W makes the three pairings collapse, which is why this term is the easy one.

10.3 The two cubic diagrams, with indices

Return to the enumeration of §8.4 and read off the contractions.

- **Sunset**, the six pairings with all three edges crossing. Each of A's indices contracts with one of B's. The six pairings are the 3! bijections between $\{i,j,k\}$ and $\{l,m,n\}$, and by symmetry of V all six give
$$V_{ijk}V_{lmn}\,G_{il}G_{jm}G_{kn}. \tag{40}$$
- **Spectacles**, the nine pairings with one edge crossing. One pair inside A, one inside B, the leftovers forced to cross. An internal pair contracts two of V's indices with a propagator, producing the *tadpole vector*
$$\boxed{u_i \;:=\; V_{ijk}\,G_{jk}} \tag{41}$$
at each end, so all nine give
$$u_i\,G_{il}\,u_l. \tag{42}$$

Multiplying by the prefactor $\frac{1}{2}(\frac{1}{3!})^2=\frac{1}{72}$ as in (28),
$$\tfrac{1}{2}\langle X_3^2\rangle_0 = \frac{1}{\beta}\Big[\tfrac{1}{12}V_{ijk}V_{lmn}G_{il}G_{jm}G_{kn} + \tfrac{1}{8}\,u_iG_{il}u_l\Big]. \tag{43}$$

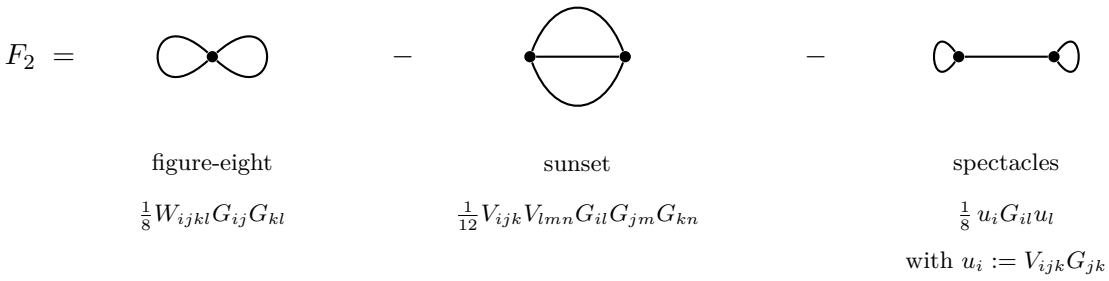


Figure 4: The two-loop coefficient, diagram by diagram. Each line is a propagator $G_{ij} = [\boldsymbol{H}_*^{-1}]_{ij}$, which is also the approximate posterior covariance. Filled dots are vertices: three lines for V_{ijk}, four for W_{ijkl}. The one-loop ring has no vertices, which is why variational Laplace never sees V or W.

Proposition 1 (Two-loop correction)**.** *With propagator $\boldsymbol{G}=\boldsymbol{H}_*^{-1}$, vertices V_{ijk} and W_{ijkl} from (37), and tadpole vector $u_i = V_{ijk}G_{jk}$,*
$$F_2 \;=\; \underbrace{\tfrac{1}{8}\,W_{ijkl}G_{ij}G_{kl}}_{\textit{figure-eight}} \;-\; \underbrace{\tfrac{1}{12}\,V_{ijk}V_{lmn}G_{il}G_{jm}G_{kn}}_{\textit{sunset}} \;-\; \underbrace{\tfrac{1}{8}\,u_iG_{il}u_l}_{\textit{spectacles}}, \tag{44}$$
and the approximation through two loops is
$$-\tfrac{1}{\beta}\log Z(\beta) \;=\; A(s^*) + \tfrac{1}{2\beta}\log\det\boldsymbol{H}_* - \tfrac{K}{2\beta}\log\tfrac{2\pi}{\beta} + \tfrac{1}{\beta^2}F_2 + \mathcal{O}(\beta^{-3}), \tag{45}$$
which at $\beta=1$ is the expansion of $-\log p(o)$.

Proof. Substitute (36) into the integral and rescale to obtain (38). Write $Z = Z_0\langle e^{-X_3-X_4-\cdots}\rangle_0$ with $Z_0 = e^{-\beta A(s^*)}\beta^{-K/2}(2\pi)^{K/2}(\det\boldsymbol{H}_*)^{-1/2}$, and apply the cumulant expansion (25). By the table of §6.3, the only terms of order β^{-1} in $\log\langle\cdot\rangle_0$ are $-\langle X_4\rangle_0$ and $\frac{1}{2}\langle X_3^2\rangle_0$, and the latter needs no $\langle X_3\rangle_0^2$ subtraction because $\langle X_3\rangle_0=0$ by parity. Evaluate the first by (39) and the second by the enumeration of §8.4, giving (43). Collecting and multiplying by $-1/\beta$ gives (45). On signs: $-\langle X_4\rangle_0$ enters $\log Z$ negatively and $-\log Z/\beta$ positively, while $+\frac{1}{2}\langle X_3^2\rangle_0$ does the reverse, so both cubic diagrams enter F_2 with a minus sign and the quartic with a plus. □

GOTCHA **Both cubic diagrams carry a minus sign.**

The figure-eight enters F_2 with $+$ and both cubic diagrams with $-$. The sign of F_2 therefore depends on whether the quartic or the cubic content dominates. In the Poisson example the two nearly cancel, which is why §15 has a row where the correction appears to make matters worse.

11 Variational Laplace is the first two terms

Restricting q to a Gaussian, expanding A to second order about its minimum and minimising is what the name variational Laplace denotes. Carrying that out returns the first two terms of the expansion.

Write $q = \mathcal{N}(m, \Sigma)$ and use the quadratic expansion in (4):

$$\mathbb{E}_q[A] \;\approx\; A(s^*) + \tfrac{1}{2}\operatorname{Tr}(\boldsymbol{H}_*\Sigma), \qquad H[q] \;=\; \tfrac{1}{2}\log\det\Sigma + \tfrac{K}{2}\log(2\pi e), \tag{46}$$

with $m = s^*$ by stationarity. Minimising $\frac{1}{2}\operatorname{Tr}(\boldsymbol{H}_*\Sigma) - \frac{1}{2}\log\det\Sigma$ over Σ gives $\Sigma = \boldsymbol{H}_*^{-1}$, which is the propagator, and substituting back the factors of $K/2$ cancel:

$$\boxed{\mathcal{F}\big[\mathcal{N}(s^*, \boldsymbol{H}_*^{-1})\big] \;=\; \underbrace{A(s^*)}_{\text{tree}} \;+\; \underbrace{\tfrac{1}{2}\log\det\boldsymbol{H}_* \;-\; \tfrac{K}{2}\log 2\pi}_{\text{one loop}}\,.} \tag{47}$$

The number variational Laplace reports is therefore the first two terms of the series, and F_2 of Proposition 1 is the next.

The Gaussian is written inside the bracket because s^* and $\boldsymbol{H}_*$ on the right are the optimised q: its mean is the mode and its covariance the inverse curvature. The optimisation over Σ is also what makes the propagator the posterior covariance rather than an inverse Hessian that merely happens to appear.

12 The one-dimensional collapse

12.1 The collapse

In one dimension $\boldsymbol{G}$, V and W are numbers, and

$$u\,G\,u \;=\; (VG)\,G\,(VG) \;=\; V^2G^3 \;=\; V_{ijk}V_{lmn}G_{il}G_{jm}G_{kn}\Big|_{K=1}. \tag{48}$$

The two contractions are the same number, so the two cubic diagrams merge into one scalar with the combined coefficient $\frac{1}{12} + \frac{1}{8} = \frac{5}{24}$, which is what §9.2 derived. Three consequences follow.

1. A one-dimensional numerical test validates the sum $5/24$ and carries no information about how it splits. Any of $(1/12, 1/8)$, $(1/8, 1/12)$, $(5/24, 0)$ and $(0, 5/24)$ passes.
2. A one-dimensional test therefore cannot catch the most likely implementation error, which is to derive $5/24$ in one dimension, take the sunset contraction as the multivariate generalisation of V^2G^3, and apply $5/24$ to it. That is wrong by the spectacles diagram, $\frac{1}{8}\big[u\boldsymbol{G}u - V_{ijk}V_{lmn}G_{il}G_{jm}G_{kn}\big]$.
3. The error puts the wrong coefficient on β^{-2}, so the residual cannot fall below that order however large β becomes. It changes the convergence rate rather than a constant.

Separating the two coefficients requires $\|u\| \neq 0$, which requires a multivariate problem with a non-vanishing tadpole vector.

12.2 An exact two-dimensional test

Take

$$A(x) = \tfrac{1}{2} x^\top \boldsymbol{H} x + \tfrac{a}{6} x_1^3 + \tfrac{b}{6} x_2^3 + \tfrac{c}{24} x_1^4 + \tfrac{d}{24} x_2^4, \qquad x \in \mathbb{R}^2, \tag{49}$$

so that $x^* = \mathbf{0}$ and the tensors can be read off: $V_{111} = a$, $V_{222} = b$ and every other entry of V zero; $W_{1111} = c$, $W_{2222} = d$ and every other entry of W zero. The tadpole vector is $u_1 = aG_{11}$, $u_2 = bG_{22}$. An off-diagonal H_{12} makes $G_{12} \neq 0$, and then

$$u\boldsymbol{G}u \;\neq\; V_{ijk} V_{lmn} G_{il} G_{jm} G_{kn}, \tag{50}$$

so the two coefficients become separately testable. Ground truth is two-dimensional adaptive quadrature, with no Monte Carlo error.

NUMBERS **Exercise 5: separating the coefficients.**

"2L correct" uses (44); "2L naive" uses $\frac{1}{8} W : \boldsymbol{GG} - \frac{5}{24} VV : \boldsymbol{GGG}$, the one-dimensional formula misapplied.

Case A: $(a, b, c, d) = (1.2, -0.9, 1.5, 1.2)$ and

$$\boldsymbol{H} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}, \qquad \boldsymbol{G} = \begin{pmatrix} 1.3333 & -0.6667 \\ -0.6667 & 1.3333 \end{pmatrix}, \qquad u = (+1.6,\, -1.2), \quad \|u\| = 2.0,$$

giving figure-eight $= +0.600000$, sunset $= -0.497778$, spectacles $= -0.986667$, so

$$F_2^{\text{correct}} = -0.884444, \qquad F_2^{\text{naive}} = -0.644444, \qquad \text{gap} = 0.240000,$$

and the gap equals $\frac{1}{8}[u\boldsymbol{G}u - VV : \boldsymbol{GGG}]$ exactly.

β	err 1L	err naive	err correct	$\text{err}_{\text{naive}} \times \beta^2$	$\text{err}_{\text{corr}} \times \beta^3$
5	$1.91 \cdot 10^{-2}$	$6.73 \cdot 10^{-3}$	$1.63 \cdot 10^{-2}$	0.168	2.04
10	$7.29 \cdot 10^{-3}$	$8.50 \cdot 10^{-4}$	$1.55 \cdot 10^{-3}$	0.085	1.55
20	$2.12 \cdot 10^{-3}$	$5.13 \cdot 10^{-4}$	$8.68 \cdot 10^{-5}$	0.205	0.69
50	$3.53 \cdot 10^{-4}$	$9.53 \cdot 10^{-5}$	$6.62 \cdot 10^{-7}$	0.238	0.08
100	$8.85 \cdot 10^{-5}$	$2.41 \cdot 10^{-5}$	$7.53 \cdot 10^{-8}$	0.241	0.08
200	$2.21 \cdot 10^{-5}$	$6.02 \cdot 10^{-6}$	$1.78 \cdot 10^{-8}$	0.241	0.14

Case B: $(a, b, c, d) = (1.5, 1.0, 2.0, 1.5)$ with $\boldsymbol{H} = \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 1.1 \end{smallmatrix}\right)$, giving $u = (+2.2297, +1.3514)$, $\|u\| = 2.607$, $F_2^{\text{correct}} = -0.414930$, $F_2^{\text{naive}} = -0.825814$ and gap $= 0.410884$. At $\beta = 200$: $\text{err}_{\text{naive}} = 1.04 \cdot 10^{-5}$ against $\text{err}_{\text{corr}} = 1.58 \cdot 10^{-7}$, a factor of 66; and $\text{err}_{\text{naive}} \times \beta^2 = 0.417$, converging to the gap 0.411.

Reading the table. Compare the scalings rather than the raw errors at small β, where both formulae are outside the asymptotic regime and either may look better by accident. The naive error times β^2 converges to the gap, so its β^{-2} coefficient is wrong and its error can never fall below $\mathcal{O}(\beta^{-2})$. The correct error times β^3 stays bounded, so its β^{-2} coefficient is right and its error is $\mathcal{O}(\beta^{-3})$. That distinction is qualitative, and it is what establishes the split $6 + 9$ rather than merely being consistent with it.

GOTCHA **Check that the saddle is the global minimum.**

Exercise 5 runs a multi-start minimisation of (49) before trusting any comparison, and reports the lowest value found ($+2.0 \cdot 10^{-15}$ and $+1.7 \cdot 10^{-14}$, so $x^* = \mathbf{0}$ is global in both cases). With a cubic term and a weak quartic, (49) can develop a second, deeper minimum, and the exact Z is then dominated by a saddle the expansion knows nothing about. Loop

corrections around the wrong saddle do not converge to the right answer. The analogue in a real inference problem is a multimodal posterior, treated in §20.

WHITEBOARD DRILL **The drill for the multivariate case.**

Draw the three two-loop diagrams. For each, write the contraction and the coefficient, and derive the coefficient by stating how many of the 15, or of the 3, pairings produce it and what the vertex prefactor is. Then design, from scratch, the smallest numerical experiment that could distinguish 1/12 from 1/8, and say which quantity must be non-zero for it to work.

13 Diagrams beyond the evidence: the shift of the mean

Every diagram so far has been a vacuum diagram contributing to $\log Z$. The same machinery corrects expectations, and the simplest case answers a question practitioners ask: whether the posterior mean sits at the mode.

At tree and one-loop level it does, since the Gaussian around s^* is symmetric and $\langle \boldsymbol{\eta} \rangle = \mathbf{0}$. The first correction comes from the cubic vertex. To leading order,

$$\langle \xi_i \rangle = \frac{\langle \xi_i e^{-X_3 - X_4 - \cdots} \rangle_0}{\langle e^{-X_3 - X_4 - \cdots} \rangle_0} = -\langle \xi_i X_3 \rangle_0 + \mathcal{O}(\beta^{-1}) = -\frac{V_{jkl}}{6\sqrt{\beta}} \langle \xi_i \xi_j \xi_k \xi_l \rangle_0 . \tag{51}$$

By (22) and the symmetry of V, the three pairings give $3V_{jkl}G_{ij}G_{kl} = 3G_{ij}u_j$, so $\langle \xi_i \rangle = -G_{ij}u_j/(2\sqrt{\beta})$, and since $\boldsymbol{\eta} = \boldsymbol{\xi}/\sqrt{\beta}$,

$$\boxed{\langle s_i \rangle - s_i^* \;=\; -\frac{1}{2\beta}\, G_{ij}u_j \;+\; \mathcal{O}(\beta^{-2})} \tag{52}$$

The diagram is a *tadpole*: a cubic vertex with two legs tied together, giving u, and the third left external, giving G_{ij}. It is the middle row of Figure 2, and it is where u takes its name. The equivalent result in the statistics literature is the posterior-moment correction of Tierney and Kadane [1986] and Tierney et al. [1989b]; see also Small [2010, ch. 6].

TRANSLATION: the practical reading

Equation (52) gives the difference between the posterior mean and the MAP estimate, in magnitude and direction, from quantities already available. The shift is proportional to the tadpole vector, which is the same object that drives the spectacles diagram, so a model with large $\|u\|$ has both a large two-loop evidence correction and a biased MAP estimate. Computing $\|\boldsymbol{G}u\|/2$ and comparing it with the posterior standard deviations costs one matrix-vector product.

QUESTION **In field theory tadpoles are said to cancel. Why do they survive here?**

They are set to zero rather than cancelling. The tadpole is the one-point function, which is the first derivative of the effective action, and the true vacuum is defined by that derivative vanishing, so "tadpoles vanish" and "the expansion is about the right vacuum" are the same statement. Expanding about the classical minimum instead leaves the linear term, removed either by shifting the field order by order or by a linear counterterm. Where a $\varphi \to -\varphi$ symmetry is present they vanish identically, which is the case usually remembered. Srednicki [2007, §4.3] sets this out.

Here the expansion is about the MAP by design, because the MAP is what variational Laplace reports. The condition $A'(s^*) = 0$ removes the one-point *vertex* and not the one-

point *function*: (52) is $-\frac{1}{2}\boldsymbol{G}u \neq 0$, and it measures the gap between mode and mean. On the worked example at $o = 0$, $-\frac{1}{2}V_*G^2 = -0.115$ against an exact -0.111.

The spectacles diagram is two tadpoles joined by a propagator,

$$-\tfrac{1}{8}\,u\boldsymbol{G}u \;=\; -\tfrac{1}{2}\,\langle\boldsymbol{\eta}\rangle^{\top}\boldsymbol{H}_*\langle\boldsymbol{\eta}\rangle \;=\; -0.010446 \quad \text{at } o = 0,$$

which is the second-order cost of expanding at the mode rather than at the mean. A field-theory counterterm would remove exactly that class. Re-expanding about the posterior mean absorbs it into the shifted expansion point.

NUMBERS **Exercise 5, second part.**

For Case A of §12.2, $u = (+1.6, -1.2)$ so $-\frac{1}{2}\boldsymbol{G}u = (-1.4667, +1.3333)$, and the prediction is that quantity divided by β. Against exact quadrature:

β	pred x_1	exact x_1	ratio	pred x_2	exact x_2	ratio
10	−0.146667	−0.155584	1.061	+0.133333	+0.138715	1.040
20	−0.073333	−0.078805	1.075	+0.066667	+0.070661	1.060
50	−0.029333	−0.030433	1.037	+0.026667	+0.027491	1.031
100	−0.014667	−0.014949	1.019	+0.013333	+0.013546	1.016
200	−0.007333	−0.007404	1.010	+0.006667	+0.006720	1.008

Both ratios approach 1 like $1 + \mathcal{O}(\beta^{-1})$, confirming the coefficient $-\frac{1}{2}$.

14 Expanding over the variational parameters instead

Every number quoted so far comes from expanding over the latents. This section is the second construction, the one the IWAI paper leads with, which expands over the variational parameters instead and gives β a reading.

14.1 Two expansions, both called Laplace

Two different integrals in this subject each receive a Laplace approximation.

	Over the latents	**Over the variational parameters**
integration variable	latent state s	parameters $\boldsymbol{\phi}$ of q
action	$A(s) = -\log p(s, o)$	$\mathcal{F}[\boldsymbol{\phi}]$
partition function	$p(o)$, exactly the evidence	$Z(\beta)$, auxiliary
the expansion gives	$-\log p(o)$, order by order	$-\beta^{-1}\log Z$, order by order
used in	§4–§16, and wherever a number is quoted	§14, §17

The two are structurally identical: the same Gaussian integral, the same log-determinant, the same diagrams, so one treatment covers both. Everything up to this point has run on the first, where Z is the evidence itself and every quoted number has a direct reading. The rest of this section develops the second. When reading a formula anywhere in this subject, check which variable is being integrated.

14.2 The construction

Take $\mathcal{F}[\boldsymbol{\phi}]$ of (4) as the action and build a partition function over $\boldsymbol{\phi} \in \mathbb{R}^N$:

$$Z(\beta) \;=\; \int_{\mathbb{R}^N} e^{-\beta\,\mathcal{F}[\boldsymbol{\phi}]}\,d^N\boldsymbol{\phi}, \qquad \beta > 0. \tag{53}$$

As $\beta \to \infty$ the integrand concentrates on $\boldsymbol{\phi}^*$ and $-\beta^{-1} \log Z \to \mathcal{F}[\boldsymbol{\phi}^*]$, so β controls how tightly an ensemble of beliefs sits on the optimum. Proposition 1 applies with $\mathcal{F}$ in place of A and $\boldsymbol{\phi}$ in place of s.

An idealised agent locates $\boldsymbol{\phi}^*$ exactly. A real one holds a distribution of beliefs about beliefs, spread around $\boldsymbol{\phi}^*$ with width $\sim \beta^{-1/2}$. On that reading $-\beta^{-1} \log Z(\beta)$ is the free energy achieved by an imperfect inference process, and the loop corrections quantify the cost of the imperfection. The ensemble is implementable: Friston [2008b] propagates a cloud of particles down a changing free-energy landscape, each performing stochastic gradient descent on $\mathcal{F}$ and dispersed about the mode by fluctuations whose amplitude sets β; read as message passing it is a particle filter [Dauwels, 2007], and the relation between ensembles and variational Bayes is studied in Wild et al. [2023]. Other readings of β are the effective sample size of a Monte Carlo scheme, the size of a neural population coding $\boldsymbol{\phi}$, and the inverse gradient-noise variance of stochastic variational inference [Hoffman et al., 2013, Kingma and Welling, 2014].

TRANSLATION: the physics name

Expanding over the parameters of the variational family rather than over the latents gives the *effective action*: a free energy as a function of the expectation value of a field, stationary at the physical configuration. Zinn-Justin [2002, ch. 9] and Altland and Simons [2010, ch. 5] develop it.

14.3 Choosing β

QUESTION **What is β, and does it correspond to anything in active inference?**

There is no standard name for it in this field, and the two constructs it is likely to be confused with are different knobs. Precision in policy selection is an inverse temperature over policies, inside the softmax on expected free energy. Cold and power posteriors temper the likelihood, $p(o \mid s)^\gamma$, which changes $\mathcal{F}$ itself [Wenzel et al., 2020]. The β of (53) leaves $\mathcal{F}$ untouched and acts one layer out.

No choice is required. At $\beta = 1$ the expansion is in the inverse curvature at the saddle, so the small quantity is the width of the posterior, and hence the amount of data. That is the Bernstein–von Mises statement [van der Vaart, 2000, ch. 10]. β is kept explicit because it makes the loop counting of §6.3 mechanical. The answer to "how much data before Laplace is safe" is to compute the three diagrams and compare them with $\frac{1}{2} \log\det \boldsymbol{H}_*$, which is the diagnostic of §18.

The structure of the terms says the same thing. Restoring β,

$$\frac{1}{\beta^2} F_2 = \frac{1}{\beta^2} \Big[\tfrac{1}{8} W : \boldsymbol{G}\boldsymbol{G} - \tfrac{1}{12} V V : \boldsymbol{G}\boldsymbol{G}\boldsymbol{G} - \tfrac{1}{8} u \boldsymbol{G} u \Big], \tag{54}$$

and each appearance of β^{-1} comes with two or three further powers of $\boldsymbol{G} = \boldsymbol{H}_*^{-1}$. The dimensionless expansion parameter is therefore

$$\epsilon \sim \frac{|W| \, \|\boldsymbol{G}\|^2}{\beta} \quad \text{and} \quad \frac{|V|^2 \, \|\boldsymbol{G}\|^3}{\beta}, \tag{55}$$

which is the scaling law to use. At $\beta = 1$ the smallness comes entirely from $\|\boldsymbol{G}\|$.

14.4 Two Boltzmann layers

	Inner layer	Outer layer
integration variable	latent state s	variational parameters $\boldsymbol{\phi}$
energy	$-\log p(s, o)$	$\mathcal{F}[\boldsymbol{\phi}]$
temperature	1, fixed by probability	β^{-1}, a modelling choice
partition function	$p(o)$, the evidence	$Z(\beta)$, auxiliary
free energy	$-\log p(o)$	$-\beta^{-1} \log Z(\beta)$

$\mathcal{F}$ is already the free energy of the inner problem over s, and (53) places a second Boltzmann layer on top of it.

One technical point. The Gaussian integral in (53) produces $\beta^{-N/2}$ from the measure, so $F_1(\beta) = \frac{1}{2} \log\det \boldsymbol{H}_* - \frac{N}{2} \log(2\pi/\beta)$ contains a $\log \beta$ and is not a β-independent coefficient. This is a standard feature of asymptotic expansions with a β-dependent measure [Bender and Orszag, 1999, §3] and it is cosmetic: redefining the measure as $\mu_\beta(d\boldsymbol{\phi}) := (\beta/2\pi)^{N/2} d\boldsymbol{\phi}$ removes it. All higher coefficients are β-independent in either convention. The unrescaled form is kept because $-\frac{N}{2} \log 2\pi$ is written out in the variational Laplace literature [Friston et al., 2007].

GOTCHA **β is not the temperature of the generative model.**

Tempering the likelihood, as in cold posteriors [Wenzel et al., 2020] or power posteriors, changes the inner layer and hence $\mathcal{F}$. The β of (53) acts entirely in the outer layer. The two interact, since loop corrections scale as $\beta^{-(L-1)}$ and tempering suppresses them, but an argument that moves between the layers without saying so is wrong.

WHITEBOARD DRILL **Both constructions.**

Write the two-layer table from memory. Then answer: on doubling the number of data points, which of $\mathcal{F}[\boldsymbol{\phi}^*]$, $\boldsymbol{H}_*$, $\boldsymbol{G}$, F_1 and F_2 grows, which shrinks, and at what rate? For i.i.d. data $\boldsymbol{H}_*$ grows like n, so $\boldsymbol{G} \sim 1/n$; count powers of $\boldsymbol{G}$ in each diagram and compare with $F_1 \sim \frac{1}{2} \log\det \boldsymbol{H}_*$, which grows like $\log n$.

FURTHER READING: Part III: tensors, cumulants, and higher-order Laplace

McCullagh [1987, ch. 2–4] is the reference for the index bookkeeping of §10: symmetric arrays of derivatives, cumulant tensors, and the contraction conventions, written for statisticians and without diagrams. Reading it alongside §8.4 is the clearest way to see that the two notations describe the same operations.

For the corrections themselves, Tierney and Kadane [1986] is the foundational paper on Laplace for posterior moments and Tierney et al. [1989b] the fully-exponential refinement that is equivalent to Proposition 1 on the evidence side. Small [2010, ch. 6] and Barndorff-Nielsen and Cox [1989, ch. 3, ch. 6] are the book-length treatments of the same expansions, the latter being the statistician's counterpart to Bender and Orszag [1999] and using entirely different vocabulary for the same objects. Butler [2007, ch. 2–3] develops the saddlepoint version with worked applications. Shun and McCullagh [1995] handles the regime where the dimension grows with the sample size.

On symmetry factors and the completeness argument of §8.5, Mattuck [1992, ch. 4] and Peskin and Schroeder [1995, §4.4]. Zinn-Justin [2002, ch. 9] has the generating-functional machinery for higher orders.

For the effective action of §14, Altland and Simons [2010, ch. 5] is the most readable derivation and Zinn-Justin [2002, ch. 9] the systematic one. Opper and Saad [2001] is an edited volume collecting the mean-field, TAP and expectation-propagation literature, which is the closest existing body of work on perturbative corrections to a variational free energy outside field theory.

Part IV
Three settings, and their numbers

15 One parameter: the Poisson–Gaussian sweep

The model is the one of §4; what is new is the sweep over the observed count, which is Table 1 of the paper. The three approximations are

$$\widehat{\mathcal{L}}_{\text{tree}} = A(s^*), \tag{56}$$

$$\widehat{\mathcal{L}}_{\text{1-loop}} = A(s^*) + \tfrac{1}{2}\log H_* - \tfrac{1}{2}\log 2\pi, \tag{57}$$

$$\widehat{\mathcal{L}}_{\text{2-loop}} = \widehat{\mathcal{L}}_{\text{1-loop}} + \tfrac{1}{8}W_* G^2 - \tfrac{5}{24}V_*^2 G^3, \tag{58}$$

with s^*, H_*, V_*, W_* and G from (10) and (11), and the exact value from adaptive Gauss–Kronrod quadrature of $e^{-A(s)}$.

NUMBERS **Exercise 6 and `two_loop_validation.py`: Table 1 of the paper, in nats.**

o	s^*	H_*	tree	1-loop	2-loop	exact	err 1L / err 2L
0	−0.5671	1.5671	1.6469	0.9526	0.9641	0.9630	0.01038 / $1.08 \cdot 10^{-3}$
1	0.0000	2.0000	1.9189	1.3466	1.3518	1.3515	0.00491 / $2.99 \cdot 10^{-4}$
2	0.4429	2.5571	2.3816	1.9321	1.9316	1.9319	0.00016 / $2.88 \cdot 10^{-4}$
3	0.7921	3.2079	2.8561	2.5200	2.5161	2.5165	0.00348 / $4.67 \cdot 10^{-4}$
5	1.3066	4.6934	3.7206	3.5748	3.5682	3.5686	0.00617 / $3.64 \cdot 10^{-4}$
10	2.0706	8.9294	5.3906	5.5664	5.5604	5.5605	0.00586 / $1.03 \cdot 10^{-4}$
20	2.8424	18.158	7.6031	8.1337	8.1299	8.1299	0.00372 / $1.66 \cdot 10^{-5}$
50	3.8323	47.168	11.2936	12.3015	12.2998	12.2999	0.00164 / $1.1 \cdot 10^{-6}$

The two-loop term removes between one and three orders of magnitude of error at every o except $o = 2$. The ratio $\log_{10}(\text{err}_{1L}/\text{err}_{2L})$ runs 0.98, 1.22, −0.27, 0.87, 1.23, 1.75, 2.35, 3.17 across the eight rows. The improvement grows with o because H_* grows like o and G shrinks, which is the mechanism of §6.3.

GOTCHA **$o = 2$ is not a counterexample.**

At $o = 2$ the one-loop error, $1.6 \cdot 10^{-4}$, is smaller than the two-loop error, $2.9 \cdot 10^{-4}$. What happens there is that F_2 passes through zero near $o = 2$, since the figure-eight and the cubic diagrams have opposite signs and cancel somewhere in $1 < o < 3$. The one-loop error then lands in the range of the next order, F_3, which is not small by comparison. The neighbours $o = 1$ and $o = 3$ show the uninterrupted pattern. Ratios of errors carry no information at an accidental zero of either error, so sweep a parameter rather than testing a point.

GOTCHA **Recompute the endpoints of a range before quoting it.**

Summarising this table as "one to four decades of improvement" is wrong at both ends. The largest reduction is 3.2 decades, not 4: the 4 comes from comparing $1.1 \cdot 10^{-6}$ with $1.6 \cdot 10^{-3}$ without taking the ratio. The smallest is −0.27, at $o = 2$, not 1.

16 Many parameters: logistic regression

This is the multivariate validation: the test that measures $\frac{1}{12}$ and $\frac{1}{8}$ separately on a model that was not constructed for the purpose.

16.1 Model, action, and cumulants

Let $\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_K)$ be a coefficient vector and $y_n \mid \boldsymbol{\gamma} \sim \text{Bernoulli}\big(\sigma(\mathbf{x}_n^\top \boldsymbol{\gamma})\big)$ for $n = 1, \dots, n_{\text{obs}}$, with $\sigma(z) = 1/(1+e^{-z})$ the logistic function, $\mathbf{x}_n$ the nth row of the design matrix and $K = 3$ covariates, the first an intercept column of ones. The action is

$$A[\boldsymbol{\gamma}] = \tfrac{K}{2} \log 2\pi + \tfrac{1}{2} \boldsymbol{\gamma}^\top \boldsymbol{\gamma} + \sum_n \Big[\log\big(1 + e^{\mathbf{x}_n^\top \boldsymbol{\gamma}}\big) - y_n \, \mathbf{x}_n^\top \boldsymbol{\gamma} \Big], \tag{59}$$

and every derivative follows from $\sigma' = \sigma(1-\sigma)$. Writing $\sigma_n := \sigma(\mathbf{x}_n^\top \boldsymbol{\gamma})$ and evaluating at the MAP,

$$H_{kl} = \delta_{kl} + \sum_n \sigma_n (1-\sigma_n) \, x_{nk} x_{nl}, \tag{60}$$

$$V_{ijk} = \sum_n \sigma_n (1-\sigma_n)(1-2\sigma_n) \, x_{ni} x_{nj} x_{nk}, \tag{61}$$

$$W_{ijkl} = \sum_n \sigma_n (1-\sigma_n)\big[1 - 6\sigma_n(1-\sigma_n)\big] x_{ni} x_{nj} x_{nk} x_{nl}. \tag{62}$$

TRANSLATION: why the brackets are cumulants

The bracketed factors are the second, third and fourth cumulants of a Bernoulli variable with success probability σ_n. The reason is that $\log(1+e^z)$ is the Bernoulli log-partition function and derivatives of a log-partition function are cumulants [Wainwright and Jordan, 2008, ch. 3]. For any exponential-family likelihood the vertices therefore have closed forms and no automatic differentiation is needed. §17 uses the same structure for the categorical case, and McCullagh [1987, ch. 2] gives the general cumulant identities.

16.2 What the third cumulant does

The third-cumulant factor $\sigma_n(1-\sigma_n)(1-2\sigma_n)$ in (61) vanishes at $\sigma_n = \frac{1}{2}$ and is antisymmetric about it. The cubic tensor, the tadpole vector and the spectacles diagram are therefore switched off when every σ_n sits at $\frac{1}{2}$.

GOTCHA **An average-case argument about a pointwise cancellation is not an argument.**

The inference that balanced labels suppress the spectacles diagram is false. Pointwise $\sigma_n = \frac{1}{2}$ is a far stronger condition than an average label rate $\bar{y} = \frac{1}{2}$. With random covariates the individual σ_n are spread over $(0,1)$ whatever the mean label rate, and the sum in (61) does not cancel. Sweeping the true intercept at $n_{\text{obs}} = 200$, averaged over 40 seeds:

intercept shift	mean $\bar{y}$	mean $\lVert V \rVert$	mean $\lVert u \rVert$
0.0	0.428	31.8	1.355
1.0	0.603	32.0	1.380
2.0	0.750	30.2	1.533
3.0	0.859	26.8	1.761
4.0	0.925	23.2	1.986

Across the sweep $\lVert u \rVert$ grows by a factor of 1.5 while $\lVert V \rVert$ decreases. Only an exactly sym-

metric design, with every covariate row appearing once at $y = 1$ and once at $y = 0$ so that $\sigma_n = \frac{1}{2}$ at the MAP, kills the cubic tensor, and then it dies completely: $\|V\| \approx 2 \cdot 10^{-7}$, $\|u\| \approx 1 \cdot 10^{-8}$. To claim that a regime suppresses a diagram, compute the tensor norm in that regime.

16.3 Ground truth

Importance sampling from the Laplace posterior is the obvious ground truth and is not accurate enough here. Since $K = 3$ the evidence integral is three-dimensional and can be computed exactly: whiten by the Laplace covariance, $\boldsymbol{\gamma} = \boldsymbol{\gamma}^* + L\boldsymbol{z}$ with $LL^\top = \boldsymbol{H}_*^{-1}$, which leaves a smooth integrand against a Gaussian weight, and apply tensor-product Gauss–Hermite quadrature. The result is stable to $\sim 10^{-13}$ nats across 30, 45 and 60 nodes per axis.

GOTCHA **A ground truth as noisy as the effect is not a ground truth.**

At $M = 5 \cdot 10^5$ draws the importance-sampling estimator has a delta-method standard error of 2.9–8.8$\cdot 10^{-4}$ nats on these eight datasets, and the two-loop residuals to be measured are $5 \cdot 10^{-5}$ to $3 \cdot 10^{-3}$ nats. The third decimal of the "exact" column would be noise and the smallest residuals would be unmeasurable.

Checking the effective sample size does not rescue this. The ESS fractions are 0.83, 0.89, 0.89, 0.96, 0.79, 0.88, 0.72, 0.81, which are healthy, and they still leave a standard error too large for the measurement. A good ESS reports that the estimator is working, not that it is precise enough for a given question. Compute the ground truth's own error bar first and check that it is an order of magnitude below the effect.

NUMBERS **Table 2 of the paper, with exact ground truth.**

logistic_exact_quadrature.py.

n_{obs}	$\bar{y}$	$\|u\|$	tree	1-loop	2-loop	exact (quad.)	$\|\Delta_{1L}\|$	$\|\Delta_{2L}\|$	ratio
True intercept 0									
20	0.20	1.02	13.059	12.129	12.071	12.074168	0.0548	0.00304	18
50	0.50	1.19	27.980	28.079	28.043	28.043325	0.0354	0.00062	57
100	0.73	1.43	47.201	48.256	48.225	48.225474	0.0301	0.00058	52
200	0.62	1.32	107.002	109.288	109.273	109.273301	0.0149	0.00005	299
True intercept +3									
20	0.90	1.19	10.318	9.072	9.039	9.038371	0.0340	0.00079	43
50	0.84	1.08	19.717	19.361	19.326	19.328332	0.0331	0.00227	15
100	0.96	2.17	22.034	22.229	22.195	22.196001	0.0328	0.00101	33
200	0.955	2.20	38.896	39.752	39.729	39.730089	0.0222	0.00079	28

Here $|\Delta_{1L}|$ and $|\Delta_{2L}|$ are the absolute errors of the one- and two-loop values against the quadrature column, and the last column is their ratio. The improvement over one loop is 15- to 300-fold. Removing the sampling error made the measured improvement larger: a noisy ground truth floors the smallest residuals and so understates the correction.

The diagram breakdown settles 1/12 against 1/8 on this model. "F_2 naive" applies the one-dimensional 5/24 to the sunset contraction:

n_{obs}	$\|u\|$	fig-8	sunset	spect	F_2	F_2 naive	err 2L	err naive	ratio
20	1.02	−0.0109	−0.0161	−0.0309	−0.0578	−0.0510	0.00304	0.00380	1.3
50	1.19	+0.0086	−0.0132	−0.0314	−0.0361	−0.0244	0.00062	0.01099	17.6
100	1.43	+0.0112	−0.0123	−0.0296	−0.0307	−0.0195	0.00058	0.01063	18.3
200	1.32	+0.0022	−0.0049	−0.0122	−0.0149	−0.0101	0.00005	0.00476	95.3
20	1.19	+0.0451	−0.0259	−0.0524	−0.0332	−0.0197	0.00079	0.01432	18.2
50	1.08	+0.0147	−0.0150	−0.0351	−0.0354	−0.0227	0.00227	0.01037	4.6
100	2.17	+0.0682	−0.0344	−0.0675	−0.0338	−0.0179	0.00101	0.01485	14.8
200	2.20	+0.0478	−0.0237	−0.0470	−0.0230	−0.0115	0.00079	0.01065	13.5

The naive coefficient is worse in all eight runs, by factors of 1.3 to 95, in both regimes. The spectacles term exceeds the sunset throughout by roughly the ratio of their coefficients, $\frac{1}{8} : \frac{1}{12}$, and the figure-eight is the largest single diagram in most of the second group.

The experiment therefore establishes that the spectacles diagram is a substantial part of F_2 in every run and that the one-dimensional $5/24$ coefficient is measurably wrong throughout. The two-dimensional test of §12.2 remains the cleaner demonstration, because there the two contractions can be made to differ by an arbitrary amount by design.

GOTCHA **Name a regime from the data, not from the generator.**

The first group of runs above was generated with a true intercept of zero and has $\bar{y}$ running from 0.20 to 0.73. Calling them balanced would describe the generator rather than the data.

17 Discrete beliefs: the simplex

This setting is closest to mainstream active inference, with discrete state spaces and categorical beliefs [Da Costa et al., 2020, Heins et al., 2022, Smith et al., 2022], and it is where the expansion gives a negative result. The expansion here runs over the variational parameters (§14), since that is where a categorical belief lives.

17.1 Coordinates, and why gauge fixing is required

A belief over K states is a point $\boldsymbol{q} = (q_1, \dots, q_K)$ on the simplex. The constraint $\sum_i q_i = 1$ makes the natural coordinates dependent, so one *gauge fixes*: keep $K-1$ logits ϕ_j and set

$$q_i = \frac{e^{\phi_i}}{1 + \sum_{j<K} e^{\phi_j}} \ (i < K), \qquad q_K = \frac{1}{1 + \sum_j e^{\phi_j}}. \tag{63}$$

The log-partition function of this exponential family is $\mathcal{A}(\boldsymbol{\phi}) = \log(1 + \sum_{j<K} e^{\phi_j})$, with $\partial \mathcal{A}/\partial \phi_i = q_i$, and its derivatives generate the categorical cumulants [Wainwright and Jordan, 2008, ch. 3]. In particular

$$[\boldsymbol{H}_*]_{ij} = \frac{\partial^2 \mathcal{A}}{\partial \phi_i \partial \phi_j} = q_i^* \delta_{ij} - q_i^* q_j^*, \tag{64}$$

the Fisher information matrix of the categorical distribution [Amari, 2016, ch. 2]. Its determinant is $\det \boldsymbol{H}_* = \prod_{i=1}^{K} q_i^*$, with the product running over all K components including the eliminated one, so the one-loop term is

$$F_1^{\text{disc}} = \tfrac{1}{2} \log \det \boldsymbol{H}_* = \tfrac{1}{2} \sum_{i=1}^{K} \log q_i^*. \tag{65}$$

GOTCHA **Gauge fixing is required, not convenient.**

Keeping all K coordinates leaves the $K \times K$ Fisher matrix $\mathrm{diag}(\boldsymbol{q}) - \boldsymbol{q}\boldsymbol{q}^\top$ with the vector $\mathbf{1}$ in its kernel, so $\det = 0$, $F_1 = -\infty$, and there is no propagator. The zero mode is the constraint direction, and gauge fixing removes it. An implementation that inverts a $K \times K$ Fisher matrix fails semi-silently, returning enormous entries of $\boldsymbol{G}$ rather than an exception.

GOTCHA **The Fisher–Rao log-volume is not the Shannon entropy.**

Both $\frac{1}{2}\sum_i \log q_i$ and $-\sum_i q_i \log q_i$ are maximised at the uniform belief, and they behave differently near a corner: as any $q_i \to 0$ the log-volume diverges to $-\infty$ while the entropy tends to a finite limit. Treating them as interchangeable predicts a bounded one-loop term. At $\boldsymbol{q}^* = (0.5, 0.3, 0.2)$ the log-volume is -1.753279 and the entropy $+1.029653$; at the uniform belief, -1.647918 and $+1.098612$. Nielsen [2020] and Amari [2016, ch. 2] give the geometry.

17.2 Worked example: $\boldsymbol{q}^* = (0.5, 0.3, 0.2)$

With $K = 3$ and gauge-fixed coordinates (ϕ_1, ϕ_2), (64) gives

$$\boldsymbol{H}_* = \begin{pmatrix} 0.25 & -0.15 \\ -0.15 & 0.21 \end{pmatrix}, \quad \det \boldsymbol{H}_* = 0.03, \quad \boldsymbol{G} = \begin{pmatrix} 7.0000 & 5.0000 \\ 5.0000 & 8.3333 \end{pmatrix}, \tag{66}$$

so $F_1 = \frac{1}{2}\log 0.03 = -1.7533$. The diagonal cumulants are

$$V_{iii} = q_i^*(1-q_i^*)(1-2q_i^*), \qquad W_{iiii} = q_i^*(1-q_i^*)\big[1 - 6q_i^*(1-q_i^*)\big], \tag{67}$$

giving $V_{111} = 0$, $V_{222} = 0.084$, $W_{1111} = -0.125$ and $W_{2222} = -0.0546$.

GOTCHA **A diagonal-only contraction is a shortcut, not a result.**

Evaluating $\frac{1}{8}W_{ijkl}G_{ij}G_{kl}$ by keeping only $i = j = k = l$ gives $\frac{1}{8}[W_{1111}G_{11}^2 + W_{2222}G_{22}^2] \approx -1.24$. That is the diagonal part, and the off-diagonal part is comparable: $W_{1112} = +0.075$ and $W_{1122} = -0.045$ are the same order as the diagonal entries, and $G_{12} = 5.0$ is the same order as G_{11}. Exercise 7 computes the exact tensors:

$$\tfrac{1}{8}W : \boldsymbol{G}\boldsymbol{G} = -0.3333 \text{ (full)} \qquad \text{against} \qquad -1.2396 \text{ (diagonal only)}.$$

The full total is

$$F_2 = \underbrace{-0.3333}_{\text{fig-8}} \underbrace{-0.2778}_{\text{sunset}} \underbrace{-0.1667}_{\text{spectacles}} = -0.7778, \qquad \frac{|F_2|}{|F_1|} = 0.444.$$

The conclusion is the same either way, since two loops is still the same order as one loop, and the shortcut overstates the figure-eight by nearly a factor of four.

TRANSLATION: getting the exact tensors

Exercise 7 constructs V and W exactly, by differentiating $H_{ij} = q_i\delta_{ij} - q_iq_j$ symbolically with the single rule

$$\frac{\partial q_a}{\partial \phi_l} = q_a(\delta_{al} - q_l), \tag{68}$$

which is all that softmax coordinates require. Every derivative is a polynomial in the q_i with Kronecker deltas, so the representation is closed and exact. The script cross-checks against

finite differences of $\mathcal{A}$, agreeing to $\sim 10^{-5}$ for the fourth derivative and exactly for the Hessian. For another model, (68) with automatic differentiation is the practical route.

17.3 The boundary

NUMBERS **Exercise 7: approaching a corner, $q^* = (1 - 2\epsilon, \epsilon, \epsilon)$.**

ϵ	$\det \boldsymbol{H}_*$	F_1	F_2	$\lvert F_2/F_1 \rvert$	$\lVert u \rVert$
0.300	$3.60 \cdot 10^{-2}$	−1.6621	−0.6806	0.409	0.224
0.200	$2.40 \cdot 10^{-2}$	−1.8649	−0.8889	0.477	0.894
0.100	$8.00 \cdot 10^{-3}$	−2.4142	−1.6875	0.699	1.565
0.050	$2.25 \cdot 10^{-3}$	−3.0484	−3.3426	1.097	1.901
0.020	$3.84 \cdot 10^{-4}$	−3.9324	−8.3368	2.120	2.102
0.010	$9.80 \cdot 10^{-5}$	−4.6153	−16.6684	3.612	2.169
0.005	$2.48 \cdot 10^{-5}$	−5.3033	−33.3342	6.286	2.203

F_2 diverges like ϵ^{-1} while F_1 diverges like $\log \epsilon$, so $\lvert F_2/F_1 \rvert$ grows without bound.

For a discrete belief approaching a corner of the simplex, the loop expansion ceases to be an expansion: the correction exceeds the term it corrects. A near-deterministic belief is the normal end state of an agent that has accumulated evidence about a hidden state or a policy, so this regime is ordinary rather than exotic.

TRANSLATION: what to do instead

Three options, in increasing order of effort.

1. Compute $\lvert F_2/F_1 \rvert$ and decline to quote a Laplace evidence when it exceeds about 0.3.
2. Avoid Laplace on the simplex. For a categorical posterior with K small the evidence is a finite sum and can be computed exactly.
3. Regularise. A Dirichlet prior with concentration greater than one keeps the saddle away from the boundary and the expansion controlled; the expansion says how much concentration is needed.

A caveat on status. Discrete-state inference does not in practice use variational Laplace. Belief updating in a partially observable Markov decision process is variational message passing on the simplex [Da Costa et al., 2020, Heins et al., 2022], which is simpler, exact for the categorical factors and free of any Gaussian approximation. The example is therefore geometric: it shows that the machinery applies wherever the Hessian is a Fisher information on a constrained parameterisation, and it marks where the expansion stops holding, which bears on any softmax-parameterised quantity.

WHITEBOARD DRILL **The discrete case.**

Derive (64) from (63) using only (68). Show $\det \boldsymbol{H}_* = \prod_{i=1}^{K} q_i^*$ for $K = 3$ by direct computation. Then say in one sentence why $F_1 \to -\infty$ as $q_3 \to 0$ while the Shannon entropy does not.

FURTHER READING: Part IV: exponential families, information geometry, and discrete inference

The common structure of all three settings is that the vertices are cumulants of an exponential family. Wainwright and Jordan [2008, ch. 3] is the reference for that statement and for the convex-duality picture behind it; McCullagh [1987, ch. 2–3] gives the cumulant identities in the index

notation used here.

For the simplex, Nielsen [2020] is the right first read: sixty pages, freely available, and it builds the Fisher metric without assuming differential geometry. Amari [2016, ch. 2–3] is the standard reference for the categorical case and the Fisher–Rao geometry, and is better used for lookup than read through.

On the practice of discrete active inference, Smith et al. [2022] is the step-by-step tutorial with code, Parr et al. [2022, ch. 4–7] the book treatment, and Da Costa et al. [2020] the careful derivation of the message passing. Heins et al. [2022] is worth reading as code: it shows what a working implementation does near the simplex boundary, which is usually to add a small concentration parameter.

For the quadrature used as ground truth in §16, Bishop [2006, ch. 10] covers the Laplace-whitened Gaussian-weight setup, and Small [2010, ch. 6] places it among the other asymptotic and numerical options.

Part V
Using it

18 The evidence, with an error bar

Quoting a Laplace evidence requires s^* and $\boldsymbol{H}_*$. The tensors V and W are two further derivatives of the same action at the same point, so computing F_2 adds no sampler, no variational family and nothing to implement beyond differentiation. What it adds to the output is a second number beside the first.

18.1 How good an error bar

The claim is that $|F_2|$ estimates the magnitude of the one-loop error. On the Poisson–Gaussian problem, sweeping the prior width $\sigma \in [0.5, 3]$ and the observed count $y \in [0, 50]$, which spans more than an order of magnitude in error size, the ratio of the true one-loop error to $|F_2|$ has median 0.98 and lies in $[0.82, 1.00]$ in 19 of 20 cases (`error_bar_calibration.py`). The one exception is a point where the one-loop error passes through zero, the $o \approx 2$ phenomenon of §15, so there $|F_2|$ over-estimates the error.

That claim is separate from whether to apply the correction. A practitioner who does not need the third decimal place of a log evidence still gains a quantity to compare against whatever is being read off. Write Δ for that quantity: a log Bayes factor, or a gap between two policies.

18.2 Three readings

	Meaning	**Action**
$\lvert F_2\rvert \ll \Delta$	The evidence supports the conclusion drawn from it.	Report the evidence and the bar, and state that the Laplace approximation is adequate at the resolution required.
$\lvert F_2\rvert \sim \Delta$	The difference being read off is not resolved by the approximation.	The comparison is not supported by the numbers behind it. Obtain a better estimate of the evidence, by quadrature in low dimension (§16) or by a sampling scheme designed for evidences.
$\lvert F_2\rvert > \lvert F_1\rvert$	The expansion has left its regime.	Do not truncate: the series is asymptotic and the two-loop value can be worse than the one-loop value. Report that the regime has been exited, and the non-perturbative estimate as the answer (§20).

The second row is the case that bears on published work. The standard methods reference for this field [Zeidman et al., 2023] warns that a Gaussian posterior may be inappropriate where

the true posterior is multimodal and prescribes sampling to assess the Laplace assumption. It records the related finding that variational Bayes is systematically overconfident in its posteriors. Neither statement supplies a number that can be computed from the fit, so the current answer to "should this evidence be trusted?" is to run a sampler.

QUESTION **I need only the parameters, not the evidence. Does this matter?**

Not for that purpose. It matters for model comparison, where the value is the output: fit two models, report the difference, and the location of either minimum is irrelevant. That describes dynamic causal modelling comparison, Bayesian model reduction, and random-effects model selection over subjects [Stephan et al., 2009, Rigoux et al., 2014, Penny, 2012], where reported log Bayes factors are often a few nats.

The premise also needs one correction. The mode is not the posterior mean, and (52) quantifies the difference: -0.115 against a true -0.111 on the worked example, with the variance 0.638 against a true 0.621, an error of 2.7%. The reported posterior carries a computable correction as well as the reported evidence.

QUESTION **Why not sample?**

Three points.

Markov chain Monte Carlo samples the posterior and does not return an evidence. A log Bayes factor requires bridge sampling, thermodynamic integration or nested sampling on top; Gelman et al. [2013, ch. 7] and Murphy [2023, ch. 11–12] survey the options.

Below roughly five to eight dimensions, quadrature is cheaper and deterministic. Whitening by the Laplace covariance and applying tensor-product Gauss–Hermite converges to $\sim 10^{-13}$ nats with no variance to report, which is what §16 does. Sparse grids and quasi-Monte Carlo extend the range.

A sampler also has an error bar that has to be compared with Δ. In Table 2 of the paper, importance sampling from the Laplace posterior gave a Monte Carlo standard error of 2.9–$8.8 \cdot 10^{-4}$ nats at $M = 5 \cdot 10^5$ draws, which is the size of the two-loop residual it was being used to measure. Replacing a biased estimator by a noisy one gains nothing when the noise is the size of the effect.

GOTCHA **A small $|F_2/F_1|$ is necessary and not sufficient.**

F_2 can be small through cancellation between the figure-eight and the cubic diagrams, which is what happens at $o \approx 2$ in §15. A cancellation at second order carries no information about third order. Report the magnitudes of the individual diagrams as well as their sum, and read §20 for the failure modes that $|F_2|$ cannot detect at all.

19 Where two loops matters

The correction is appreciable when $|F_2|/|F_1|$ is not small. Since $F_1 \sim \frac{1}{2} \log\det \boldsymbol{H}_*$ grows logarithmically in the curvature while F_2 carries two or three powers of $\boldsymbol{G}$, the ratio is governed by

$$\frac{|F_2|}{|F_1|} \sim \frac{|W|\,\|\boldsymbol{G}\|^2 + |V|^2 \|\boldsymbol{G}\|^3}{\left|\log\det \boldsymbol{H}_*\right|}. \tag{69}$$

Three classes populate that regime. For each, the useful question is which tensor is responsible, since that identifies the diagram to watch.

Non-Gaussian likelihoods at modest effective precision. Here W and V are both comparable to $\boldsymbol{H}_*$. Counts, rates and binary outcomes qualify, and the Poisson and logistic ex-

amples are canonical: across those runs the two-loop term is $3.9 \cdot 10^{-3}$ to $5.7 \cdot 10^{-2}$ nats per parameter, spanning the precision at which the thresholds of Kass and Raftery [1995] begin to bite. Tierney et al. [1989b] identified the same regime from the statistics side, where the Laplace error degrades from $\mathcal{O}(n^{-2})$ to $\mathcal{O}(n^{-1})$ when the log-posterior has substantial third- and fourth-derivative content at the mode.

Hierarchical and scale-mixing models. Here V and W couple directions of very different curvature. In hierarchical models [Friston, 2008a, Penny et al., 2003, Rue et al., 2009] the eigenvalues of $\boldsymbol{H}_*$ span orders of magnitude; the loose directions dominate $\det \boldsymbol{H}_*$, while the cumulant tensors mix loose and tight directions in a way one loop cannot represent. This is the regime integrated nested Laplace [Rue et al., 2009] addresses.

Discrete posteriors near a simplex boundary. Here $\boldsymbol{G}$ diverges as $\det \boldsymbol{H}_* \to 0$, quantified in §17. This class bounds the construction rather than extending it, and is where it produces a warning rather than a refinement.

20 Limits of the construction

Four limits.

The series is asymptotic. Adding orders helps and then stops helping (§6.4). A two-loop number quoted without an estimate of the expansion parameter is incomplete, and §17 shows a regime in which the two-loop term exceeds the term it corrects. Bender and Orszag [1999, ch. 3] and Boyd [1999] are the references.

The expansion is about one minimum. If the action has several well-separated minima, the contribution of the ones not expanded about is of order $e^{-\Delta A}$ relative to the one retained, where ΔA is the difference in action. That quantity has a vanishing Taylor series in the expansion parameter, so no order of the expansion contains it and $|F_2|$ cannot detect it. A multimodal posterior therefore requires a different method: expectation propagation [Minka, 2001], a sampling scheme, or an asymptotic treatment that sums over the minima rather than expanding about one. The corresponding failure of variational Laplace is documented directly by Asadpour et al. [2026] for dynamic causal models of bistable circuits. This limit is a separate problem from the one addressed here and is treated in follow-up work.

The fourth-derivative tensor is large. W has K^4 entries. For a generalised-linear action the contractions collapse onto quantities of the size of the data rather than of K, and for a general action they can be evaluated without materialising the tensor, but neither route is implemented in the accompanying code.

Expected free energy is out of scope. Planning under expected free energy [de Vries et al., 2025, Nuijten et al., 2026, Parr and Friston, 2019] has an epistemic term that depends on beliefs the agent does not yet hold, so the objective is not a fixed function of $\boldsymbol{\phi}$ and §10 does not apply unchanged.

21 A recipe for your own model

1. **Identify the integration variable.** Either the latent s with action $-\log p(s, o)$ at $\beta = 1$, as in §15, or the variational parameters $\boldsymbol{\phi}$ with energy $\mathcal{F}[\boldsymbol{\phi}]$, as in §14. Write it down explicitly; §14.1 is the confusion that follows from skipping this.

2. **Find the minimum, and check whether it is the only one.** Multi-start if the problem is not convex. If competing modes exist, see §20 before reporting anything.

3. **Form $\boldsymbol{H}_*$ and $\boldsymbol{G} = \boldsymbol{H}_*^{-1}$**, which the pipeline already computes, and record $\det \boldsymbol{H}_*$ and the condition number.

4. **Form V_{ijk} and W_{ijkl}**, by automatic differentiation (`jax.jacfwd` composed three and four times, or `torch.autograd.functional`) or analytically if the model is an exponential family, in which case they are cumulants with closed forms as in (61)–(62) and (67). Check that V and W are invariant under permutation of their indices to numerical tolerance; if not, the derivatives are wrong.

5. **Compute the three diagrams separately.** With `numpy`:

```
fig8 = 0.125 * np.einsum("ijkl,ij,kl->", W, G, G)
sunset = (1/12) * np.einsum("ijk,lmn,il,jm,kn->", V, V, G, G, G)
u = np.einsum("ijk,jk->i", V, G)
spect = 0.125 * u @ G @ u
F2 = fig8 - sunset - spect
```

   Keep them separate in the output, as §18 requires.

6. **Compute $|F_2/F_1|$** with $F_1 = \frac{1}{2}\log\det \boldsymbol{H}_*$. Above about 0.3, report that the approximation is out of its regime rather than reporting a corrected evidence.

7. **Check a Gaussian special case.** Set the non-Gaussian part of the model to zero and confirm $V = W = 0$ and that the one-loop value matches an exact computation. Exercise 3 does this and it catches sign and factor errors immediately.

8. **Validate against quadrature in low dimension**, with at least two dimensions, a non-zero off-diagonal Hessian and non-zero $\|u\|$. §12 explains why a one-dimensional check cannot detect a multivariate error.

9. **Compute the mean shift** (52). It costs one matrix-vector product and reports whether the MAP estimate is biased.

> **GOTCHA** **Contract rather than materialise.**
>
> W_{ijkl} has K^4 entries, which is awkward for K in the hundreds. The diagrams only ever require contractions: $W : \boldsymbol{GG}$ is a scalar, and so is every other term of (44). An implementation at scale should start from the contraction and evaluate it by Hessian-vector-product-style methods, rather than forming the tensor. Daxberger et al. [2021] documents the structured and last-layer approximations used for the Hessian at neural-network scale, and the same techniques apply here.

22 All the gotchas, in one list

22.1 Conceptual

1. Two integrals in this subject each receive a Laplace approximation, one over the latents and one over the variational parameters. Know which is being expanded (§14.1).

2. The Kullback–Leibler identity is exact and does not make variational Laplace exact. Two approximations are stacked and F_2 corrects the second (§4.4).

3. A loop is a cycle in a diagram of Wick contractions, not a cycle in a factor graph. A mean-field posterior on a tree-shaped graphical model still has every loop correction discussed here (§8).
4. The series is asymptotic, so more loops is not monotonically better (§6.4).
5. β is not likelihood tempering, which is a different knob acting on $\mathcal{F}$ itself (§14).
6. Loop corrections around one minimum of a multimodal action converge to the wrong answer, and $|F_2|$ does not detect the situation (§20).

22.2 Technical

7. Wick's theorem pairs slots, not distinct indices (§7.1).
8. 5/24 is a one-dimensional coincidence. In more dimensions it splits into 1/12 and 1/8, and a one-dimensional test cannot detect the difference (§12).
9. Both cubic diagrams enter F_2 with a minus sign and the quartic with a plus (§10).
10. F_2 is often a difference of larger diagrams, so coefficient errors are amplified and a small $|F_2|$ may be a cancellation (§9, §18).
11. $\langle\cdot\rangle_0$ and $\langle\cdot\rangle$ are different measures; the propagator is an input rather than an answer (§7.4).
12. The one-loop ring carries $\frac{1}{2}\log H_*$ and comes from the measure, not from a contraction (§**??**).
13. Loop-order counting differs for $\log Z$ and for expectations with external legs (§8.5).
14. F_1 contains a $\log\beta$; F_2 and beyond do not (§14).
15. Gauge fixing on the simplex is required, or $\det \boldsymbol{H}_* = 0$ (§17).
16. The Fisher–Rao log-volume is not the Shannon entropy (§17).
17. Diagonal-only tensor contractions are shortcuts: the discrete figure-eight changes from -1.24 to -0.33 when evaluated in full (§17).
18. The Poisson saddle equation is $e^s + s = o$, not $s\,e^s = o$ (§4).
19. A pure cubic action has no convergent partition function; keep $W > 0$ (§9.2).

22.3 Empirical hygiene

These are about validating a perturbative correction against a ground truth rather than about loop expansions, and they transfer.

20. A ground truth as noisy as the effect is not a ground truth. Compute its error bar first and check that it is an order of magnitude below what is being measured; a healthy effective sample size does not settle this (§16).
21. An average-case argument about a pointwise cancellation is not an argument: $\bar{y} = 1/2$ does not give $\sigma_n = 1/2$ (§16).
22. Name a regime from the data rather than from the generator (§16).
23. Recompute the endpoints of any range before quoting it (§15).
24. Sweep a parameter rather than testing a point, so that an accidental zero of the error does not read as a failure (§15).

25. Compare scalings rather than raw errors when choosing between two candidate formulae; outside the asymptotic regime either may look better by accident (§12.2).

26. Check that the minimum found is the global one before trusting any comparison (§12.2).

23 What is new and what is not

23.1 Not new

The loop expansion is textbook [Negele and Orland, 1988, Zinn-Justin, 2002, Kardar, 2007]. Higher-order corrections to Laplace are long established in statistics [Tierney et al., 1989b, Shun and McCullagh, 1995, Small, 2010]. Organising corrections to mean-field theory by cumulants of a local field is the TAP expansion [Plefka, 1982, Georges and Yedidia, 1991] and the Bethe–Kikuchi cluster construction [Yedidia et al., 2005], collected in Opper and Saad [2001]. That variational Laplace is a saddle-point approximation is understood by everyone who uses it. At one loop there is no new mathematics here.

23.2 New

1. The multivariate decomposition and the fact that one dimension hides it: the split of the cubic contribution into sunset (1/12) and spectacles (1/8), the tadpole vector $u_i = V_{ijk}G_{jk}$ as the object that separates them, and the demonstration that a one-dimensional validation cannot detect the difference (§12).

2. A calibrated diagnostic: $|F_2|$ estimates the one-loop error to about ten per cent on the family where it was measured, and $|F_2/F_1|$ diverges for discrete beliefs approaching a simplex corner (§18, §17).

3. Numerical demonstrations throughout, including the ones that came out negative (§17).

23.3 Why the language is worth adopting

The diagrammatic organisation collapses $\mathcal{O}(K^k)$ cumulant contractions at order k into a few named topologies, so the bookkeeping does not change with K. It makes the asymptotic structure explicit, since each loop carries one more power of the expansion parameter. And it connects this literature to a body of technique — resummation families, large-K counting, uniform asymptotics, resurgence — that has not been applied to variational inference. Whether that technique pays off is open.

23.4 What would settle how much it matters

Two outcomes are possible and the difference is empirical. Either there exist model comparisons whose conclusion turns on F_2, with the corrected and uncorrected log evidences falling on opposite sides of a decision, in which case the correction belongs in the standard calculation; or $|F_2|$ sits uniformly below the Monte Carlo error of the ground truth in use, in which case the diagnostic remains useful and the correction is decoration. The natural test is behavioural model comparison in computational psychiatry, where per-subject log evidences are combined by random-effects model selection [Stephan et al., 2009, Rigoux et al., 2014] and reported log Bayes factors are a few nats. Active-inference generative models for the two-step task [Daw et al., 2011] already exist [Gijsen et al., 2022, Garrido-Pérez et al., 2026], so the correction could be added to a fitted pipeline.

24 Relation to other higher-order schemes

Fully exponential Laplace. Tierney et al. [1989b,a] compute higher-order corrections to posterior expectations and marginal densities by exploiting the fully exponential form of the integrand. On the evidence side their construction is equivalent to Proposition 1, and their posterior-expectation results recover (52) and its higher analogues. The two agree numerically; the diagrammatic version has bookkeeping that does not change with the number of parameters. Small [2010, ch. 6] compares the approaches.

Expectation propagation. Minka [2001] replaces the Gaussian-around-the-mode assumption by site-wise moment matching, refining iteratively rather than perturbatively, and is presented at textbook length in Bishop [2006, ch. 10] and Murphy [2023, ch. 10]. The failure modes are complementary: expectation propagation handles multimodality, which no finite-order expansion about one minimum can, while the loop expansion gives a closed-form single-pass refinement for one mode with modest non-Gaussianity.

Integrated nested Laplace. Rue et al. [2009] combines Laplace approximations across the levels of a hierarchical latent-Gaussian model with numerical integration over hyperparameters. The two-loop expansion supplies an analytic form for the per-level corrections that are currently approximated numerically, so the direct extension is analytic two-loop at the latent-Gaussian level with numerical integration over hyperparameters.

TAP and Bethe–Kikuchi. Plefka [1982], Georges and Yedidia [1991], Yedidia et al. [2005] organise corrections to mean-field theory by cumulants of a local field, which is structurally the same idea; Opper and Saad [2001, ch. 2] and Mézard and Montanari [2009, ch. 14] give the overview. TAP is vertex-based as here, Bethe–Kikuchi is region-based, and the two agree on trees and differ on loopy graphs.

Within active inference. Koudahl et al. [2026] builds a Bethe cluster-expansion correction to the variational objective of predictive coding, reaching the same instinct from the graphical-model side. Tschantz et al. [2025] is a distinct construction by an overlapping group, maintaining posteriors over network parameters and deriving closed-form Hebbian updates, with no cluster expansion.

GOTCHA **Cite the paper, not the group.**

The two references above share six authors and have adjacent titles while doing different things. When two papers from one group sit next to each other in a reference list, check which one carries the result being attributed.

FURTHER READING: Part V: model comparison in practice, and what to read next

For the use the correction is for, Gelman et al. [2013, ch. 7] covers evidence and model comparison applied, Kass and Raftery [1995] supplies the reading scale, and Penny [2012] describes the comparison procedure as practised in this field. Stephan et al. [2009] and Rigoux et al. [2014] are the group-level random-effects schemes the correction would feed into.

For alternatives to a perturbative correction, Bishop [2006, ch. 10] and Murphy [2023, ch. 10–11] cover expectation propagation and the sampling options, and Small [2010, ch. 6] places them alongside the asymptotic ones. Daxberger et al. [2021] reports what Laplace at neural-network scale actually does, and is the reference for the structured Hessian approximations needed to apply §21 at that size.

For extending the expansion itself, Zinn-Justin [2002, ch. 9] has the generating-functional ma-

chinery for higher orders and Altland and Simons [2010, ch. 5] the effective-action formalism. Aniceto et al. [2019] covers resurgence, and Boyd [1999] should be read first as a calibration of how much resummation repays.

For the connection to graphical-model methods, Opper and Saad [2001] is the collection, Yedidia et al. [2005] the region-based construction, and Mézard and Montanari [2009, ch. 14] the statistical-physics account. The relationship between a vertex expansion and a region expansion on a loopy graph has not been worked out for the variational free energy.

25 The whiteboard test

Solutions in Appendix C.

Q1. Derive $\mathcal{F}[\boldsymbol{\phi}] = \mathrm{KL}[q_{\boldsymbol{\phi}} \| p(\cdot \mid o)] - \log p(o)$ and deduce the bound. Then say why the bound does not make variational Laplace exact.

Q2. Evaluate the Gaussian integral $\int_{\mathbb{R}^K} e^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{H}\boldsymbol{x}}\, d^K x$, and name the object that appears when its logarithm is negated.

Q3. How many Wick pairings are there for $\langle \eta^8 \rangle_0$, and what is $\langle \eta^8 \rangle_0$ in one dimension with variance G?

Q4. Why is the expansion of $\log Z$ rather than of Z?

Q5. Why is the quadratic term of the action treated differently from the others?

Q6. Starting from $A(x) = \frac{1}{2}Hx^2 + \frac{1}{6}Vx^3 + \frac{1}{24}Wx^4$, derive F_1 and F_2.

Q7. Which cumulants contribute at two loops, and why does no other term survive? At what order does the fifth derivative first appear?

Q8. Enumerate the 15 pairings of two cubic vertices, split them, and show that no third topology exists.

Q9. Where does 5/24 come from, and why is it wrong in more than one dimension?

Q10. Design the smallest numerical experiment that distinguishes 1/12 from 1/8. Which quantity must be non-zero, and what is compared?

Q11. For the Poisson–Gaussian model at $o = 1$, compute s^*, H_*, G and F_2 exactly.

Q12. Show that $L = E - V_{\mathrm{tot}} + 1$ and $3V_3 + 4V_4 = 2E$ give $\frac{1}{2}V_3 + V_4 = L - 1$. What are the two-loop solutions, and why does that make Figure 3 complete?

Q13. Derive the shift of the mean, (52), including the factor $-\frac{1}{2}$. What does the spectacles diagram equal in terms of it?

Q14. On the three-state simplex, why is $\det \boldsymbol{H}_*$ a product over all three q_i^* when there are only two coordinates? What happens without gauge fixing?

Q15. A colleague reports a log evidence with $|F_2| = 10^{-4}$ nats and concludes that the Laplace approximation is safe to three decimal places. Give two distinct ways that conclusion can fail, and the check for each.

A Notation

Every symbol is also defined where it first appears. This table is the backstop.

Symbol	Meaning
s	Hidden (latent, unobserved) state; a real number in §4–§9, a vector in $\mathbb{R}^K$ from §10 onward, a categorical belief in §17.
K	$\dim s$.
o	Observed data, held fixed throughout.
$p(s, o)$	Generative model: the joint density of latents and data.
$p(o)$	Model evidence (marginal likelihood), $\int p(s, o)\, ds$.
$p(s \mid o)$	Exact posterior, $p(s, o)/p(o)$.
$A(s)$	The *action*, $-\log p(s, o)$: everything in the exponent of (1).
s^*	The *saddle*: the minimiser of A, that is, the MAP estimate.
η, $\boldsymbol{\eta}$	Fluctuation about the saddle, $s = s^* + \eta$.
H_*, $\boldsymbol{H}_*$	Second derivative of A at s^*; a number when $K = 1$, a matrix otherwise.
G, $\boldsymbol{G}$	The *propagator*, $\boldsymbol{G} = \boldsymbol{H}_*^{-1}$, which is also the approximate posterior covariance.
V_*, V_{ijk}	Third derivative of A at s^*: the *cubic vertex*.
W_*, W_{ijkl}	Fourth derivative of A at s^*: the *quartic vertex*.
u_i	*Tadpole vector*, $u_i = V_{ijk} G_{jk}$.
X	Everything the quadratic approximation discards, Eq. (14); X_3 and X_4 are its cubic and quartic parts, Eq. (18).
$\langle\cdot\rangle_0$	Expectation under the quadratic term alone, that is, under $\mathcal{N}(\mathbf{0}, \boldsymbol{G})$ with the vertices switched off, Eq. (16).
$\langle\cdot\rangle$	Expectation under the exact posterior. The subscript distinguishes them throughout; see §7.4.
F_1	One-loop coefficient, $\frac{1}{2}\log\det \boldsymbol{H}_* - \frac{K}{2}\log 2\pi$: the correction variational Laplace applies.
F_2	Two-loop coefficient, Eq. (2).
F_L	Coefficient at loop order L.
L	Loop order: the number of independent cycles in a diagram, §8.5.
$\mathcal{S}$	Symmetry factor of a diagram; the diagram's value is divided by it, §8.3.
β	Bookkeeping parameter that orders the expansion, Eq. (18). Set to 1 except in §14.
Z, $Z(\beta)$	Partition function. Expanding over the latents, $Z = p(o)$ exactly; in §14 it is an integral over variational parameters.
$q_{\boldsymbol{\phi}}(s)$, $\boldsymbol{\phi}$	Variational posterior and its parameters, $\boldsymbol{\phi} \in \mathbb{R}^N$ with $N = \dim \boldsymbol{\phi}$. Used in §3, and in §14 and §17 where the expansion runs over $\boldsymbol{\phi}$.
$\mathcal{F}[\boldsymbol{\phi}]$	Variational free energy, Eq. (4); an upper bound on $-\log p(o)$.
$\mathrm{KL}[q\|p]$	Kullback–Leibler divergence, $\mathbb{E}_q[\log q - \log p] \geq 0$.
$\sigma(z)$	Logistic function $1/(1 + e^{-z})$ (§16).
q_i^*	Components of a categorical belief at the saddle (§17).
Δ	The quantity being read off the evidence: a log Bayes factor, or a gap between two policies (§18).
nat	Unit of log evidence, in natural-log units; 1 nat $\approx$ 1.44 bits. A log Bayes factor of 3 nats is conventionally read as strong evidence [Kass and Raftery, 1995].
Einstein sum	Repeated indices are summed over $1, \ldots, K$ unless stated otherwise.

GOTCHA **Three sign conventions are in circulation.**

We minimise A and expand $-\log Z$, so every quantity here is on the energy side: smaller is better, and $-\log p(o)$ is the surprise. Much of the machine-learning literature works with the ELBO, $-\mathcal{F}$, and maximises it [Blei et al., 2017]. The physics literature works with $+\log Z$. Check which convention a source uses before concluding that a sign is wrong.

B A dictionary between the two vocabularies

The right-hand column names an object defined in this primer; the left-hand column is the term to search for in the physics literature. With the identification of the action with $-\log p(s, o)$,

the manipulations are the same manipulations.

B.1 The table

Statistical field theory	Variational inference	Comment
Euclidean action $S[\varphi]$	the action $A(s) = -\log p(s, o)$, or $\mathcal{F}[\boldsymbol{\phi}]$	The quantity in the exponent that is minimised.
field φ	latents s, or variational parameters $\boldsymbol{\phi}$	A zero-dimensional field: finitely many components, no space or time.
$\hbar$, or temperature	β^{-1}	§14; the entry most likely to mislead.
partition function Z	model evidence $p(o)$	Expanding over the latents, the two are the same integral.
free energy $-\log Z$	surprise $-\log p(o)$	What model comparison consumes.
classical, or tree level	the MAP value $A(s^*)$	The first term.
one loop	Occam factor $\frac{1}{2}\log\det \boldsymbol{H}_*$	The second term, and all of variational Laplace.
propagator G_{ij}	posterior covariance $[\boldsymbol{H}_*^{-1}]_{ij}$	The same matrix under two names.
n-point vertex	nth derivative of the action at the saddle	V_{ijk} and W_{ijkl} suffice at two loops.
vacuum diagram	a term in the evidence expansion	No external legs, so it contributes to $\log Z$ rather than to a correlation function (§**??**).
external leg	an averaged fluctuation	The number of legs is the quantity being computed.
tadpole $u_i = V_{ijk}G_{jk}$	the shift of the posterior mean	§13.
effective action	expansion over the variational parameters	§14: a free energy as a function of a belief.
loop counting	order in the asymptotic series	L loops carries one more power of the expansion parameter than $L-1$.
Wick's theorem	Isserlis' theorem; Gaussian moment identities	The same identity in two literatures.
linked-cluster theorem	cumulant expansion of $\log Z$	Why only connected diagrams appear.
symmetry factor $\lvert\mathrm{Aut}\rvert$	the rational coefficient of a contraction	8, 12 and 8 for the three diagrams here.
anharmonic oscillator	non-Gaussian posterior	The factorial growth of the former is why the latter's expansion is asymptotic.
instanton, e^{-S}	a second posterior mode, $e^{-\Delta A}$	Invisible at every order of the expansion (§20).

B.2 How much of the apparatus is in play

With the vocabulary in hand, here is how much of the machinery those words carry with them.

The field theory used here is *zero-dimensional*: the field is a finite list of numbers at a single point rather than a function on space or time. A loop is therefore a power of a matrix, or a sum over a finite index, rather than an integral over momenta, and every diagram is a finite sum of finite numbers. Four consequences follow, and they remove most of what a physics text spends its length on.

- **No oscillatory integrals.** The integrand is a probability density. There is no factor of i, no stationary phase and nothing to rotate. The saddle is a genuine minimum, $\boldsymbol{H}_*$ is positive definite there, and every propagator is real. An instinct to worry about contours can be switched off.
- **Nothing diverges.** A zero-dimensional theory with finitely many components has no divergent integrals, so no regularisation and no renormalisation. Every quantity in this document is a finite sum of finite numbers.
- **No thermodynamic limit.** K counts parameters rather than a spatial volume. There is no extensivity to exploit and diagrams do not simplify at large K unless the model supplies extra structure.
- **The expansion parameter is a precision, not a coupling.** It is small because a curvature is large, which is to say because there is a lot of data (§6.3). Weak coupling here means a sharp posterior, and the regime practitioners care about is the one where the parameter is not small.

What does carry over is the bookkeeping, and it carries over exactly: Wick's theorem, the linked-cluster theorem, symmetry factors and loop counting are the same in any number of dimensions, and only the cost of evaluating a loop changes. A reader opening one of the books in the reading lists should expect most of its length to concern the apparatus listed above; the shared material is Mattuck [1992, ch. 3–4] and Negele and Orland [1988, ch. 2], and Kardar [2007, ch. 2–3] treats it with no factors of i anywhere.

Restoring a dimension, by letting the field depend on time, turns a loop from a sum into an integral and the propagator from a matrix into a function of a time difference. That is a different construction and is not treated here.

GOTCHA **One entry of the dictionary is asymmetric.**

In field theory the action is given and the measure is flat. Expanding over the variational parameters, $\mathcal{F}$ is itself the free energy of an inner Boltzmann problem over the latent states, and (53) adds a second Boltzmann layer over $\boldsymbol{\phi}$: two layers, two notions of temperature (§14). Expanding over the latents, as this primer does throughout, there is one layer and the issue does not arise.

C Solutions to the whiteboard test

Q1. Insert $p(s, o) = p(s \mid o)p(o)$ into (4); the $\log p(o)$ is constant under $\mathbb{E}_q$ and comes out, leaving $\mathbb{E}_q[\log q - \log p(s \mid o)] = \mathrm{KL} \geq 0$, so $\mathcal{F} \geq -\log p(o)$ with equality when $q_{\boldsymbol{\phi}} = p(\cdot \mid o)$. The bound does not make variational Laplace exact because two approximations are stacked: restricting q to a family, whose error is the Kullback–Leibler gap, and expanding the action to second order about the mode, whose error is the truncation error of an asymptotic series. F_2 is the next term of the second. The arithmetic demonstration is that the reported value 0.9526 lies below the exact 0.9630, which a bound cannot (§4.4).

Q2. $(2\pi)^{K/2}(\det \boldsymbol{H})^{-1/2}$. Diagonalise $\boldsymbol{H}$ by an orthogonal transformation, which has unit Jacobian, and apply the one-dimensional result once per eigenvalue, using $\prod_a \lambda_a = \det \boldsymbol{H}$. Negating the logarithm produces $+\frac{1}{2}\log\det \boldsymbol{H}$, the Occam factor.

Q3. $7!! = 105$ pairings, so $\langle \eta^8 \rangle_0 = 105\, G^4$.

Q4. Because $\log Z$ is the sum of connected diagrams only (the linked-cluster theorem, (26)). Expanding Z produces disconnected diagrams that the cumulant subtractions in $\log Z$ remove exactly, and $\log Z$ is the quantity wanted in any case.

Q5. Because it is the one term that can be integrated in closed form, so it goes into the measure and the rest into the integrand. It is absorbed into the Gaussian that the remaining terms are averaged against, and what survives of it is its inverse, $G = \langle \eta^2 \rangle_0$, which relates two fluctuations and is therefore drawn as a line. The power counting agrees: under the rescaling of (18) the nth term carries $\beta^{1-n/2}$, and $n = 2$ is the only vanishing exponent (§6.2).

Q6. See §9.2: rescale $x = \xi/\sqrt{\beta}$, keep $-\langle X_4 \rangle_0 + \frac{1}{2}\langle X_3^2 \rangle_0$, use $\langle \xi^4 \rangle_0 = 3G^2$ and $\langle \xi^6 \rangle_0 = 15G^3$, giving $F_1 = \frac{1}{2}\log H - \frac{1}{2}\log(2\pi/\beta)$ and $F_2 = \frac{1}{8}WG^2 - \frac{5}{24}V^2G^3$.

Q7. $\langle X_4 \rangle_0$ and $\frac{1}{2}\langle X_3^2 \rangle_0$. Odd moments vanish by parity, removing $\langle X_3 \rangle_0$, $\langle X_3 X_4 \rangle_0$ and $\langle X_5 \rangle_0$; every remaining candidate carries β^{-2} or smaller by the vertex counting $\beta^{-(n-2)/2}$. The fifth derivative first appears at three loops, via $\langle X_3 X_5 \rangle_0$, which is what keeps a two-loop implementation cheap (§6.3).

Q8. See §8.4: six with all three edges crossing, the sunset; nine with one self-pair per vertex and one crossing edge, the spectacles; fifteen in total. No third topology exists because a cubic vertex has three legs, an odd number, so the crossing count is $3 - 2p \in \{3, 1\}$ for $p \in \{0, 1\}$ self-pairs, and a self-pairing in one vertex forces one in the other.

Q9. From $\langle \xi^6 \rangle_0 = 15G^3$ and the prefactor $1/72$, $15/72 = 5/24$. In more than one dimension the fifteen pairings give two different tensor contractions, six giving $VV : \boldsymbol{GGG}$ and nine giving $u\boldsymbol{G}u$, which coincide only when $K = 1$. Applying $5/24$ to the sunset contraction alone is wrong by $\frac{1}{8}[u\boldsymbol{G}u - VV : \boldsymbol{GGG}]$.

Q10. Two dimensions suffice. Required: $\|u\| \neq 0$, so a non-zero cubic tensor, and $G_{12} \neq 0$, so an off-diagonal Hessian, giving $u\boldsymbol{G}u \neq VV : \boldsymbol{GGG}$. Compare the β-scaling of the residuals rather than the raw errors: the wrong formula's error times β^2 tends to the gap between the two contractions, while the right formula's error times β^3 stays bounded (§12.2).

Q11. $s^* = 0$, since $e^0 + 0 = 1$; then $H_* = 2$, $G = \frac{1}{2}$, $V_* = W_* = 1$, and $F_2 = \frac{1}{8} \cdot \frac{1}{4} - \frac{5}{24} \cdot \frac{1}{8} = \frac{1}{32} - \frac{5}{192} = \frac{1}{192} \approx 0.005208$.

Q12. Substitute $E = (3V_3 + 4V_4)/2$ into $L = E - (V_3 + V_4) + 1$: $L = (3V_3 + 4V_4)/2 - V_3 - V_4 + 1 = V_3/2 + V_4 + 1$, hence $V_3/2 + V_4 = L - 1$. At $L = 2$ the solutions are $(V_3, V_4) = (2, 0)$, giving the sunset and the spectacles, and $(0, 1)$, giving the figure-eight. The list is complete because those are the only non-negative integer solutions.

Q13. $\langle \xi_i \rangle = -\langle \xi_i X_3 \rangle_0 = -(V_{jkl}/6\sqrt{\beta})\langle \xi_i \xi_j \xi_k \xi_l \rangle_0$; the three Wick pairings give $3G_{ij}u_j$ by symmetry of V, so $3/6 = 1/2$, and $\boldsymbol{\eta} = \boldsymbol{\xi}/\sqrt{\beta}$ supplies the second factor of $\beta^{-1/2}$, giving $-G_{ij}u_j/(2\beta)$. The spectacles diagram is two such tadpoles joined by a propagator, $-\frac{1}{8}u\boldsymbol{G}u = -\frac{1}{2}\langle \boldsymbol{\eta} \rangle^\top \boldsymbol{H}_* \langle \boldsymbol{\eta} \rangle$.

Q14. In gauge-fixed coordinates (ϕ_1, ϕ_2) the 2×2 matrix is $\mathrm{diag}(q_1, q_2) - \boldsymbol{q}_{1:2}\boldsymbol{q}_{1:2}^\top$, whose determinant is $q_1q_2 - q_1^2q_2 - q_1q_2^2 = q_1q_2(1 - q_1 - q_2) = q_1q_2q_3$: the eliminated component re-enters through the constraint. Without gauge fixing the 3×3 Fisher matrix annihilates $\mathbf{1}$, so $\det = 0$, $F_1 = -\infty$, and no propagator exists.

Q15. First, F_2 may be small by cancellation between the figure-eight and the cubic diagrams, which says nothing about F_3. The check is to report the three diagrams separately and to sweep a parameter rather than trusting one point (§15, §18). Second, $|F_2|$ is blind to other minima of the action: if the posterior is multimodal, the missing contribution is of order $e^{-\Delta A}$ and has a vanishing Taylor series in the expansion parameter, so no order detects it. The check is to search for other minima before reporting anything (§20).

D Gaussian integral and moment identities

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}Hx^2}dx = \sqrt{2\pi/H}, \tag{70}$$

$$\int_{\mathbb{R}^K} e^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{H}\boldsymbol{x}}d^K x = (2\pi)^{K/2}(\det \boldsymbol{H})^{-1/2}, \tag{71}$$

$$\int_{\mathbb{R}^K} e^{-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{H}\boldsymbol{x}+\boldsymbol{J}^\top \boldsymbol{x}}d^K x = (2\pi)^{K/2}(\det \boldsymbol{H})^{-1/2} e^{\frac{1}{2}\boldsymbol{J}^\top \boldsymbol{H}^{-1}\boldsymbol{J}}, \tag{72}$$

$$\langle \eta^{2m}\rangle_0 = (2m-1)!!\, G^m \qquad (1, 3, 15, 105 \text{ for } m = 1, 2, 3, 4), \tag{73}$$

$$\langle \xi_i \xi_j\rangle_0 = G_{ij}, \tag{74}$$

$$\langle \xi_i \xi_j \xi_k \xi_l\rangle_0 = G_{ij}G_{kl} + G_{ik}G_{jl} + G_{il}G_{jk}, \tag{75}$$

$$\text{number of pairings of } 2m = (2m-1)!!. \tag{76}$$

Logistic derivative: $\sigma' = \sigma(1-\sigma)$. Softmax derivative: $\partial q_a/\partial \phi_l = q_a(\delta_{al} - q_l)$. Loop counting: $\frac{1}{2}V_3 + V_4 = L - 1$. Two-loop formula: (44). Shift of the mean: (52).

E Reproducing every number

Section	Script	What it establishes
§4, §9	`two_loop_validation.py`	The five constants of the worked example at $o = 0$, and the arithmetic of the correction.
§7.1	`primer_examples.py` Ex. 1	Wick's theorem to 10^{-14} by exact Gauss–Hermite quadrature; the $(2m-1)!!$ counting.
§8.4	Ex. 2	The 15 pairings enumerated and classified $6+9+0$; the coefficients $6/72$ and $9/72$.
§6.4, §9.2	Ex. 3	One loop exact for Gaussian actions; the figure-eight coefficient alone; $\mathrm{err}_{1L}\times\beta^2 \to \lvert F_2\rvert$.
§9.2, §12	Ex. 4	The combined $5/24$ in one dimension, and the cost of using $1/12$.
§12.2, §13	Ex. 5	Separation of $1/12$ from $1/8$ against exact two-dimensional quadrature; the shift of the mean.
§15	Ex. 6, `two_loop_validation.py`	Every intermediate number at $o = 1$; Table 1 of the paper.
§17	Ex. 7	Exact categorical cumulant tensors; full against diagonal contraction; divergence of $\lvert F_2/F_1\rvert$ at the boundary.
§16	`logistic_regression_validation.py`	Table 2 of the paper by importance sampling, and why that was not accurate enough.
§16	`logistic_exact_quadrature.py`	The same table by exact quadrature, with the diagram breakdown that settles $1/12$ against $1/8$.
§18	`error_bar_calibration.py`	The error-bar calibration: median ratio 0.98 over the sweep, and 19 of 20 cases within $[0.82, 1.00]$.

All scripts are deterministic. They are published alongside this document at `https://github.com/egolabs-ai/iwai26`, under `two-loops/code`, with a `pyproject.toml` that pins the only two dependencies, `numpy` and `scipy`:

`uv sync` then `uv run python primer_examples.py`

`primer_examples.py` runs in a couple of seconds; the slowest part is the two-dimensional adaptive quadrature in Exercise 5.

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